CALIFORNIA STATE UNIVERSITY
LONG BEACH

CECS 341: Computer Architecture and Organization

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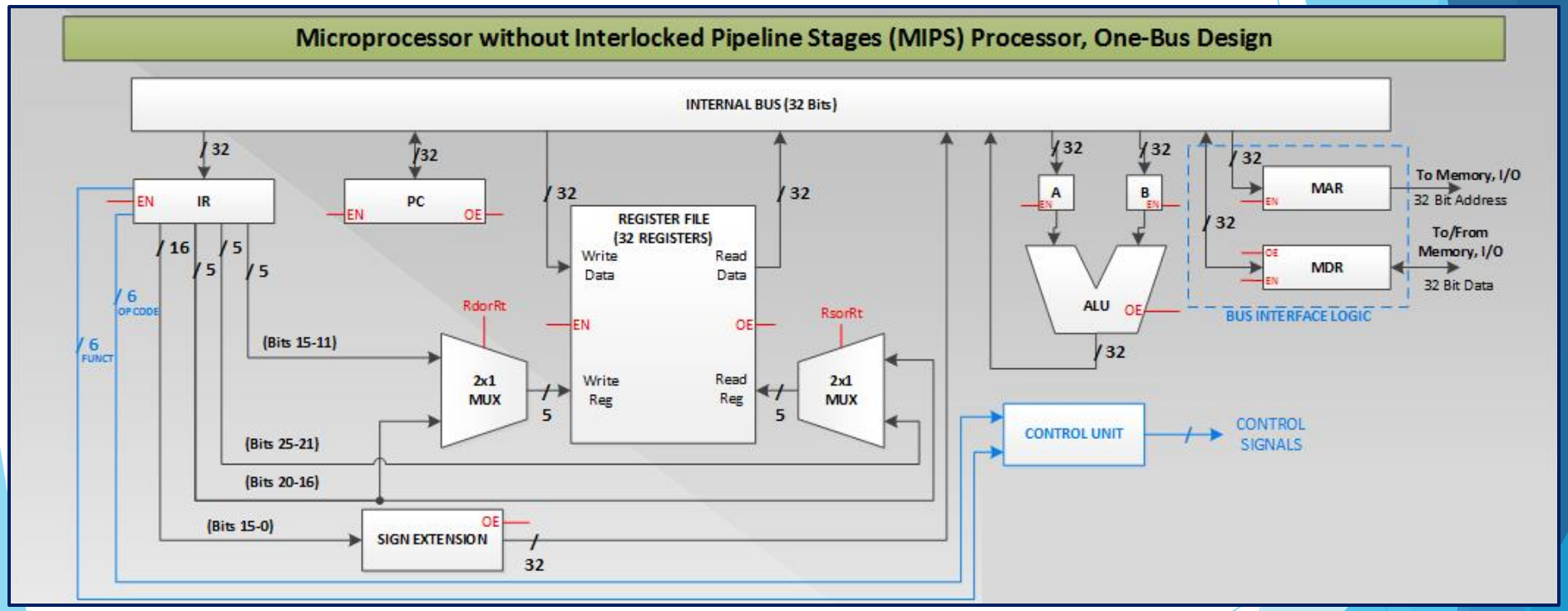
Fall 2025

Lecture Outline

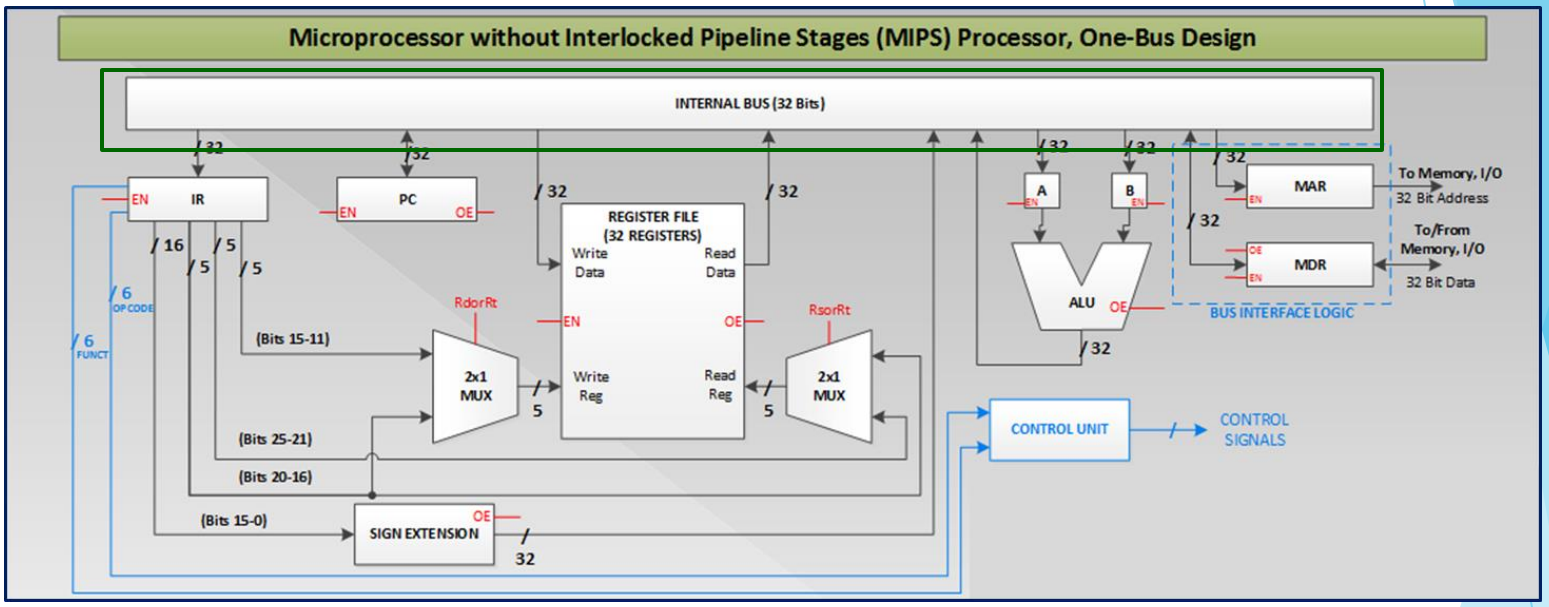
- Overview of MIPS *One-bus* implementation
  - Internal CPU bus
  - Datapath components
  - Tri-state buffers
  - Control signals
- Incremental design of MIPS Datapath

Overview of MIPS *One-bus* implementation

MIPS One-bus Organization

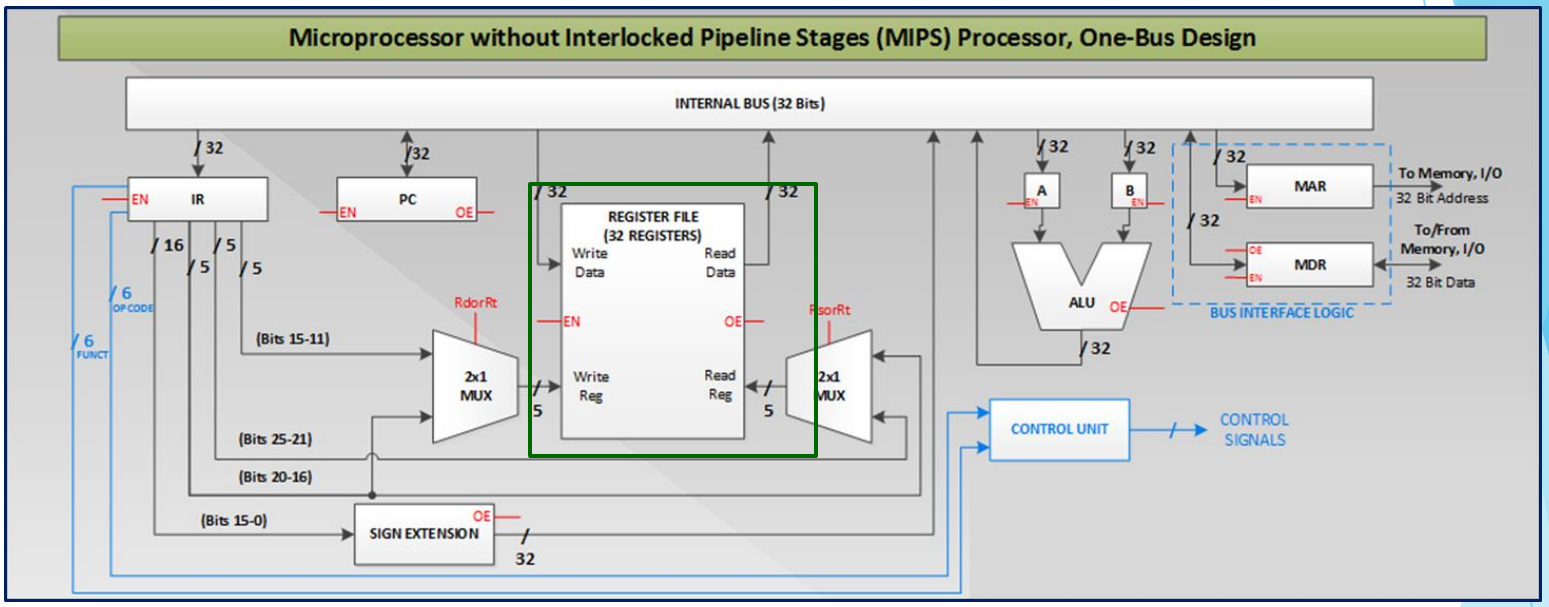


Internal Bus



- The *One-bus* organization of the MIPS architecture uses a single 32-bit bus to interconnect all of the datapath components.
  - Data is transferred between datapath components in 32-bit words.
  - Registers are 32-bits wide.
  - ALU operands are 32-bits wide.

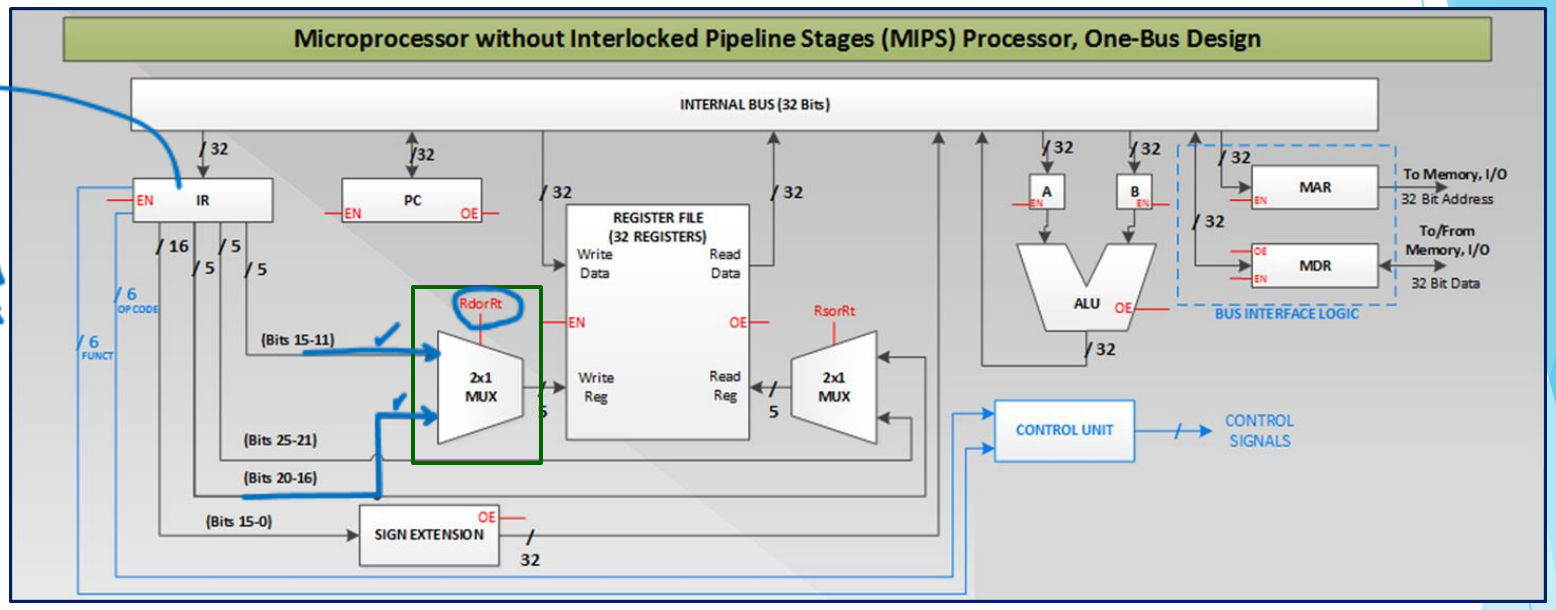
Register File



- Realizes the *register set* specified in the MIPS architecture.
  - Thirty-two 32-bit registers.
  - Two ports (read, write).

Write Register #	Read Register #
5-bit field	5-bit field
Specifies the destination register.	Specifies a source register.
Either Rt or Rd field of instruction.	Either Rs or Rt field of instruction

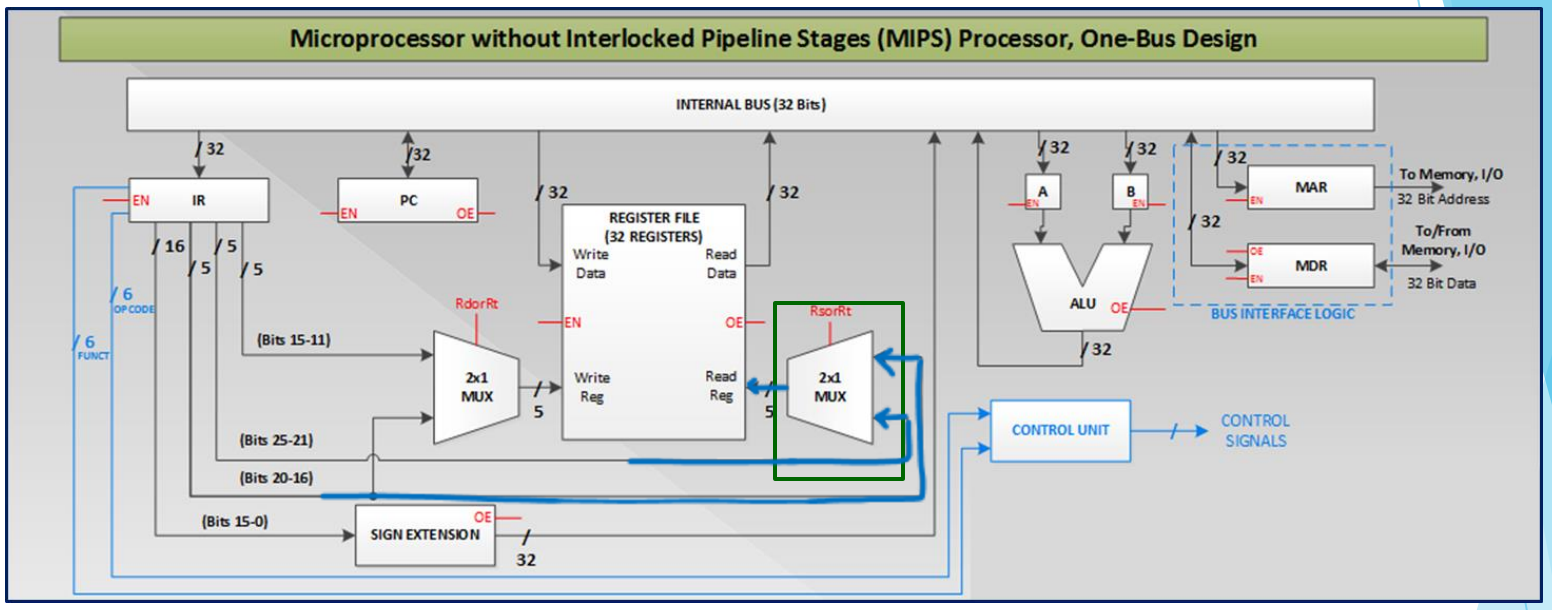
Write Register Multiplexer



- Destination register specified in **Rd** or **Rt**.
- Write register multiplexer selects the appropriate register field to specify the destination register.
  - 2-to-1 Multiplexer

Field Name	Instruction Register
Rt(4..0)	IR(20..16)
Rd(4..0)	IR(15..11)

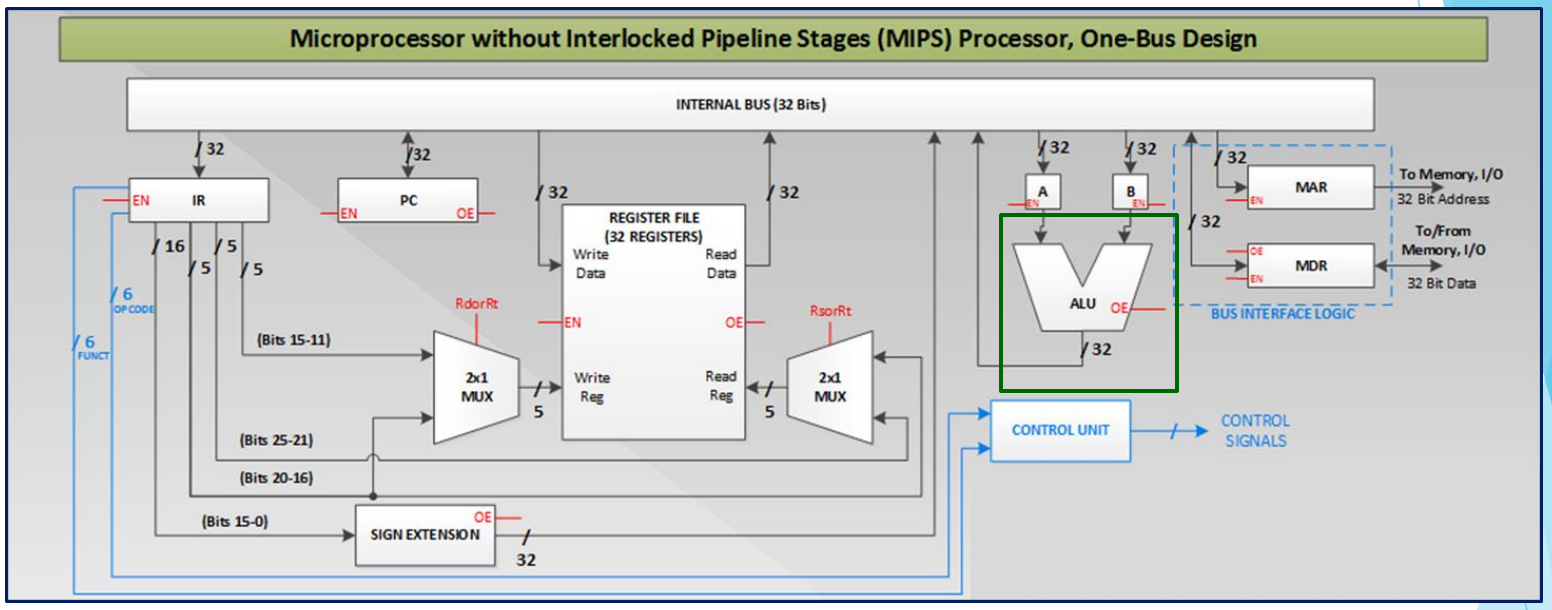
Read Register Multiplexer



- Source register specified in **Rs** or **Rt.**
- Read register multiplexer selects the appropriate register field to specify a source register.
  - 2-to-1 Multiplexer

Field Name	Instruction Register
Rs(4..0) →	IR(25..21)
Rt(4..0) →	IR(20..16)

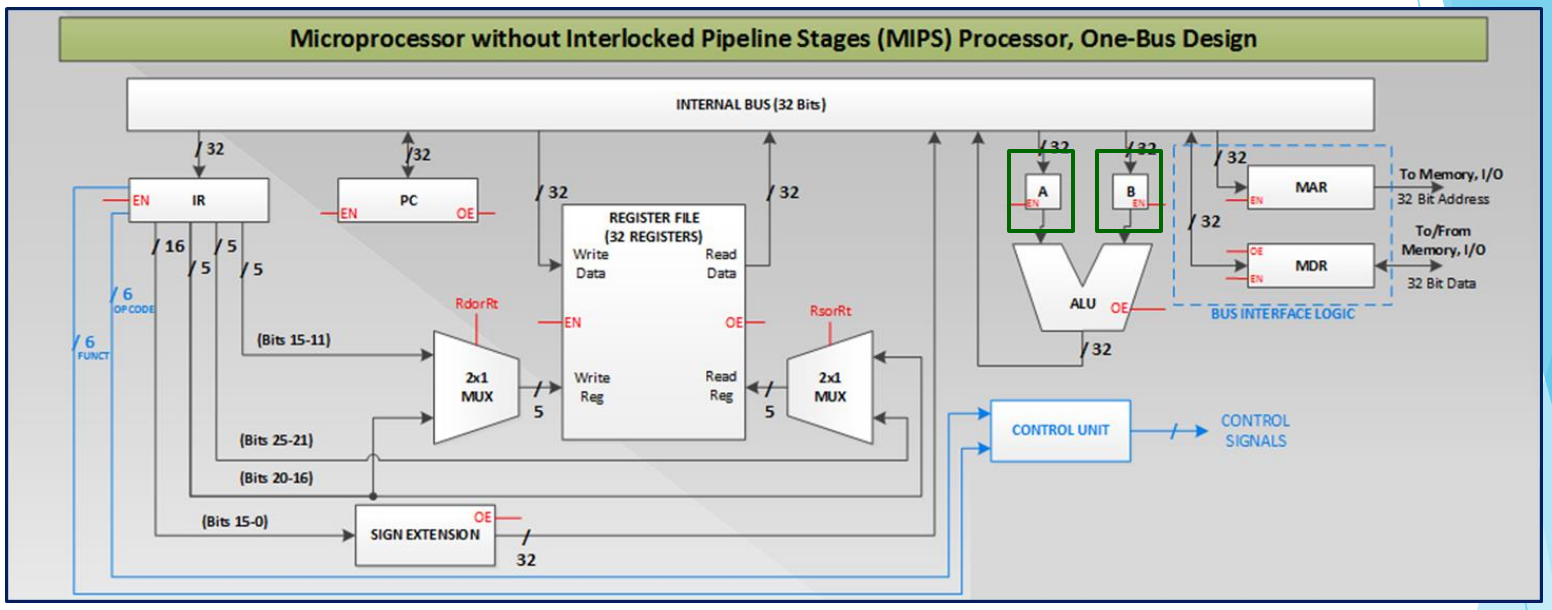
Arithmetic and Logic Unit (ALU)



- Implements arithmetic and logic operations.
- Takes two 32-bit operands and produces a 32-bit result (and overflow).
- The ALU is combinational logic.
  - Does not require a clock.
  - Does not store information.



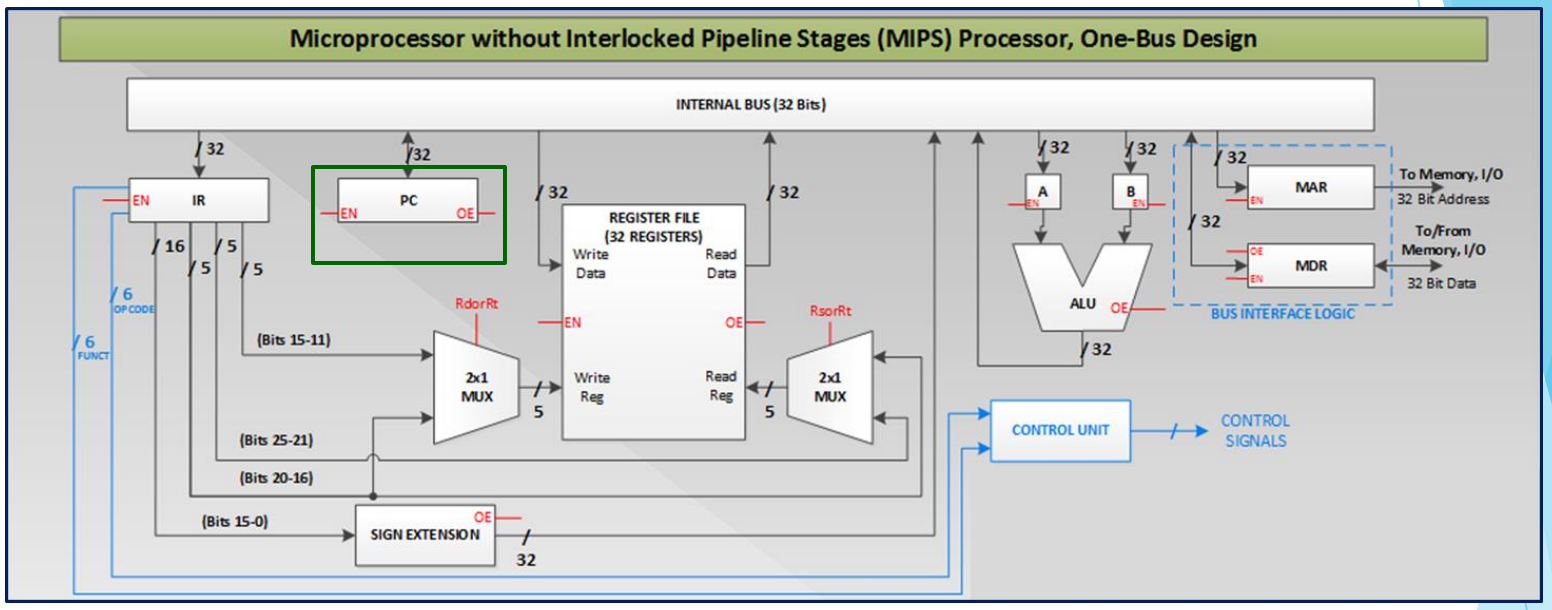
Temporary Registers



- 32-bit registers.
- Register A stores one ALU operand (input).
- Register B stores the other ALU operand (input).

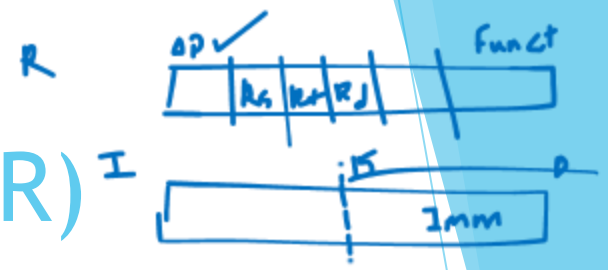
Program Counter (PC)

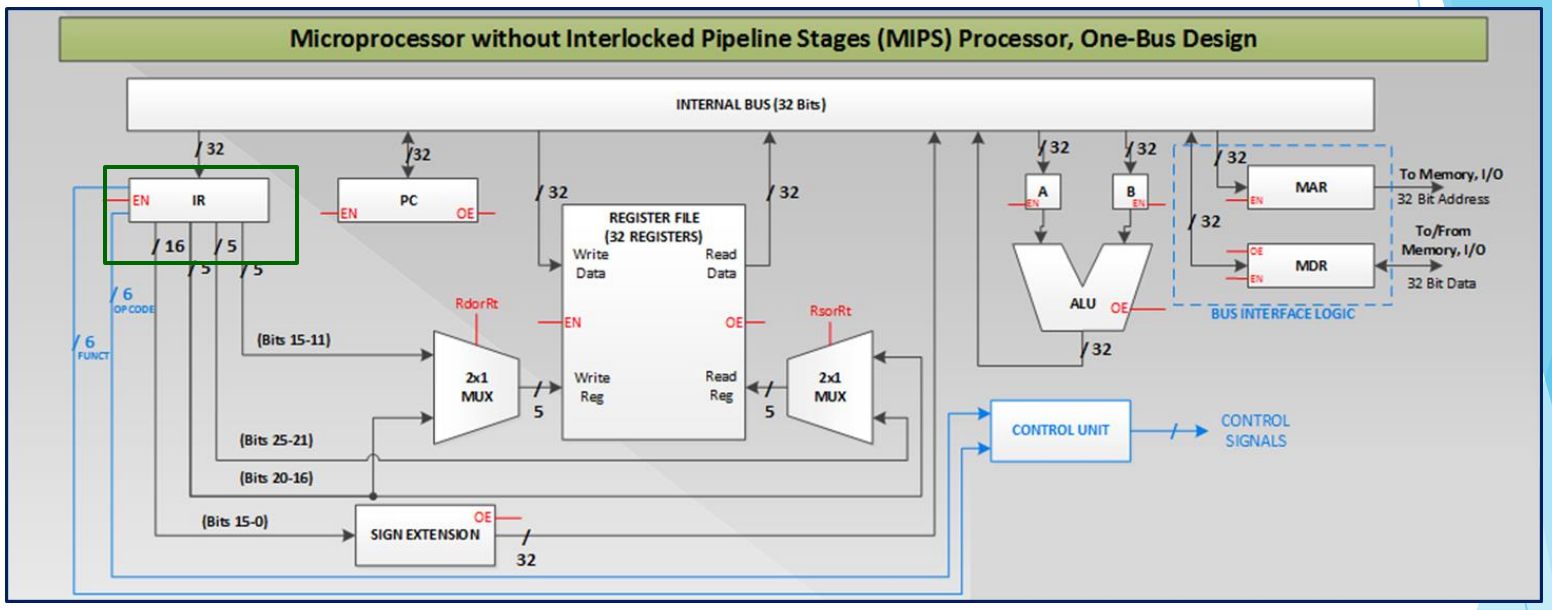
Inst Pointer



- 32-bit register.
- Stores the *address of the instruction* to be fetched from (program) memory.

Instruction Register (IR)

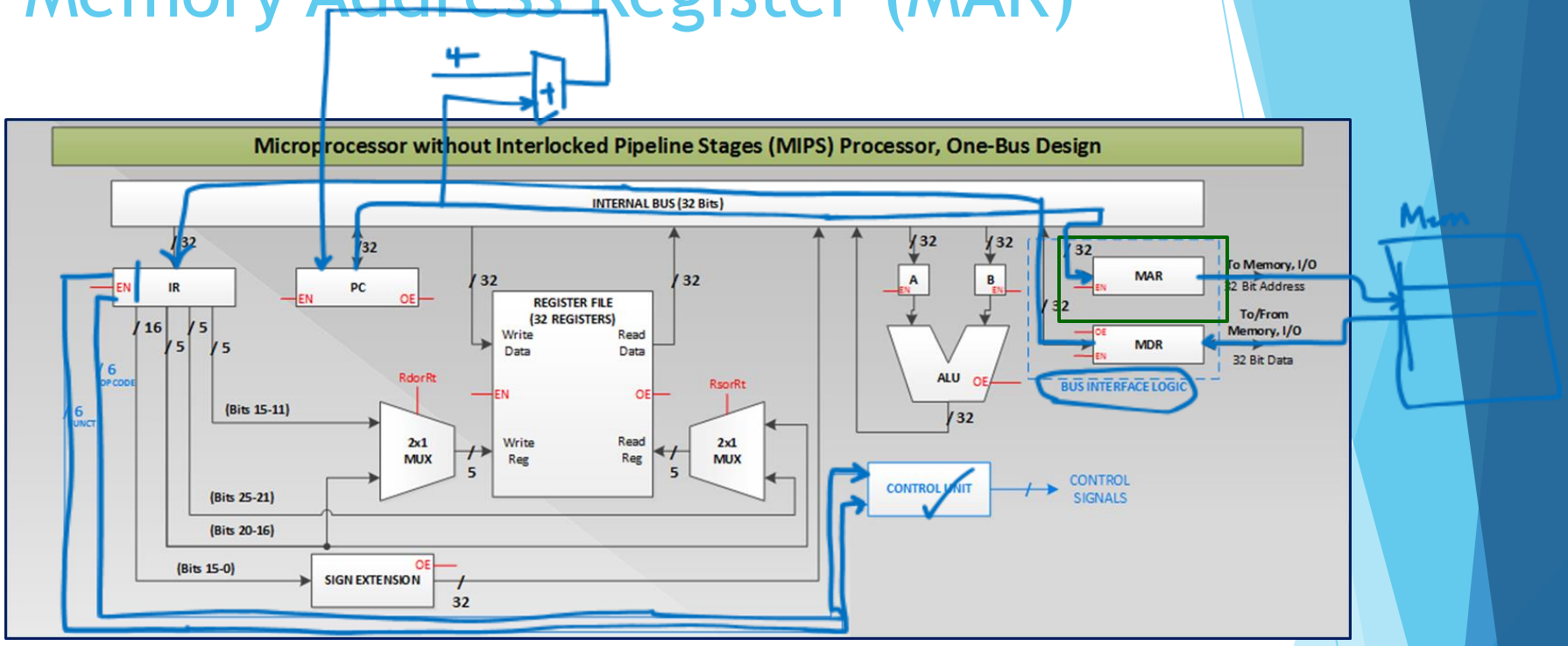




- 32-bit register.
- Stores the *instruction* being executed by the processor.
  - Previously fetched from memory.
- Realizes the IR specified in the MIPS arch.

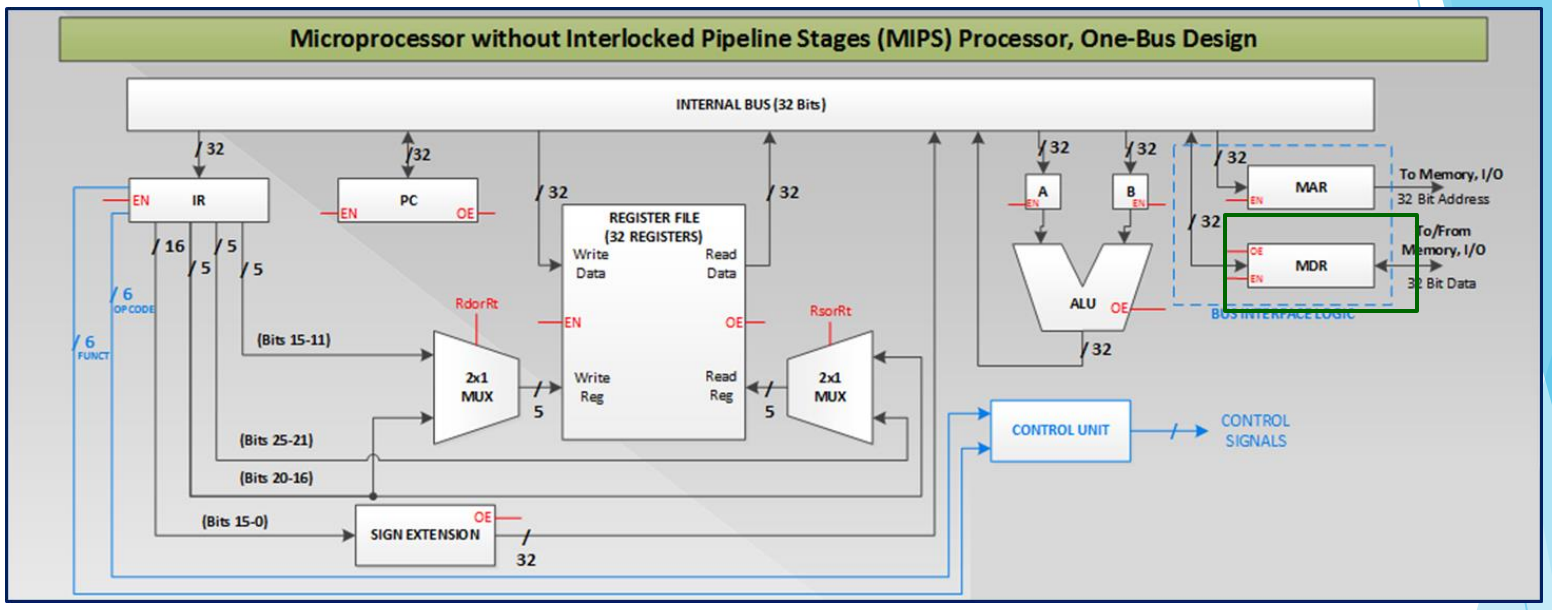
Instruction Register	Field Name
IR(31 – 26)	opcode(5 – 0)
IR(25 – 21)	Rs(4 – 0)
IR(20 – 16)	Rt(4 – 0)
IR(15 – 11)	Rd(4 – 0)
IR(15 – 0)	Imm(15 – 0)

Memory Address Register (MAR)



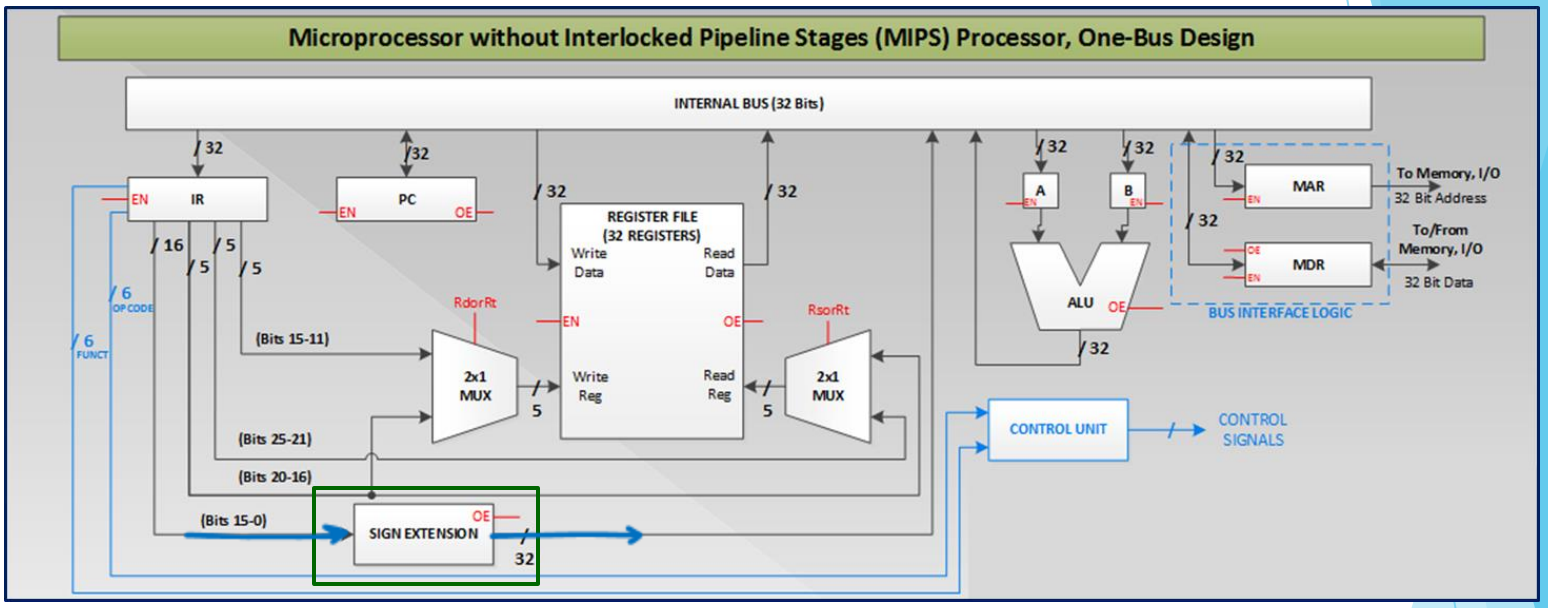
- 32-bit register.
- Stores the *address* of the data to be read from or written to (data) memory.
- Part of the *bus interface logic*.

Memory Data Register (MDR)



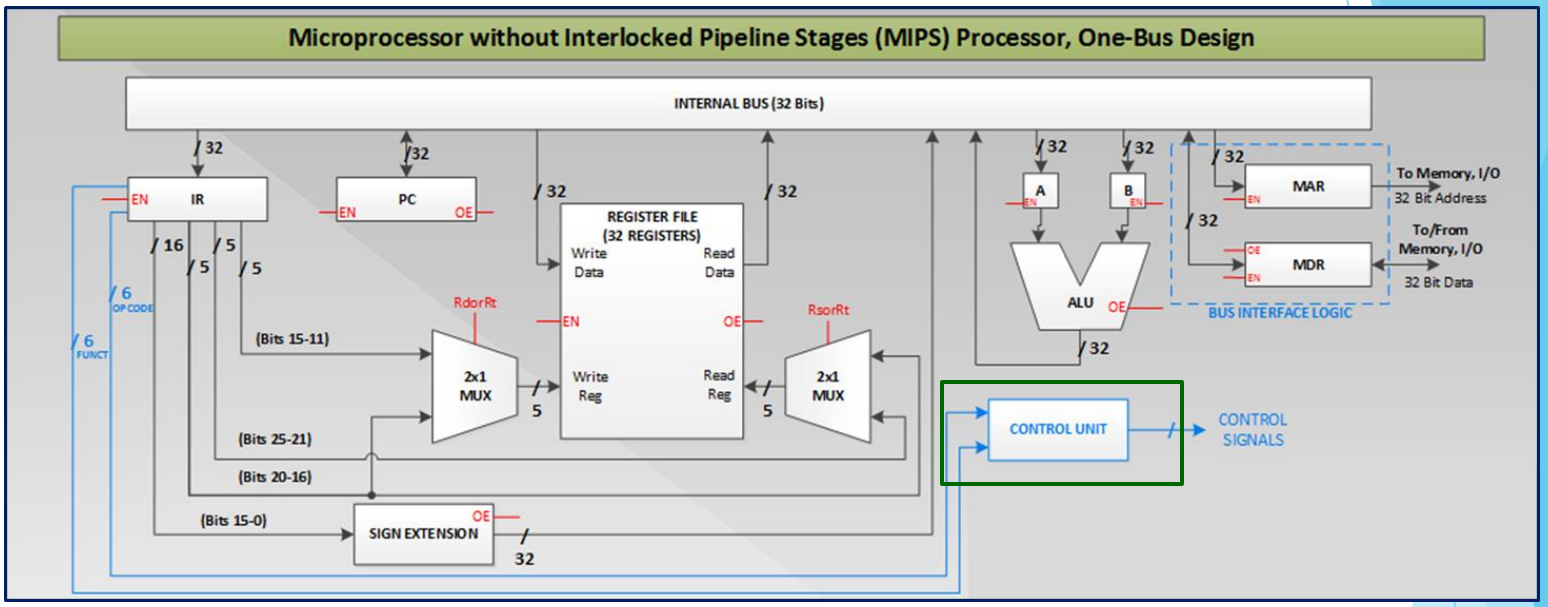
- 32-bit register.
- Stores the data read from or written to (data) memory.
- Part of the *bus interface logic*.

Sign Extension Unit (SEU)



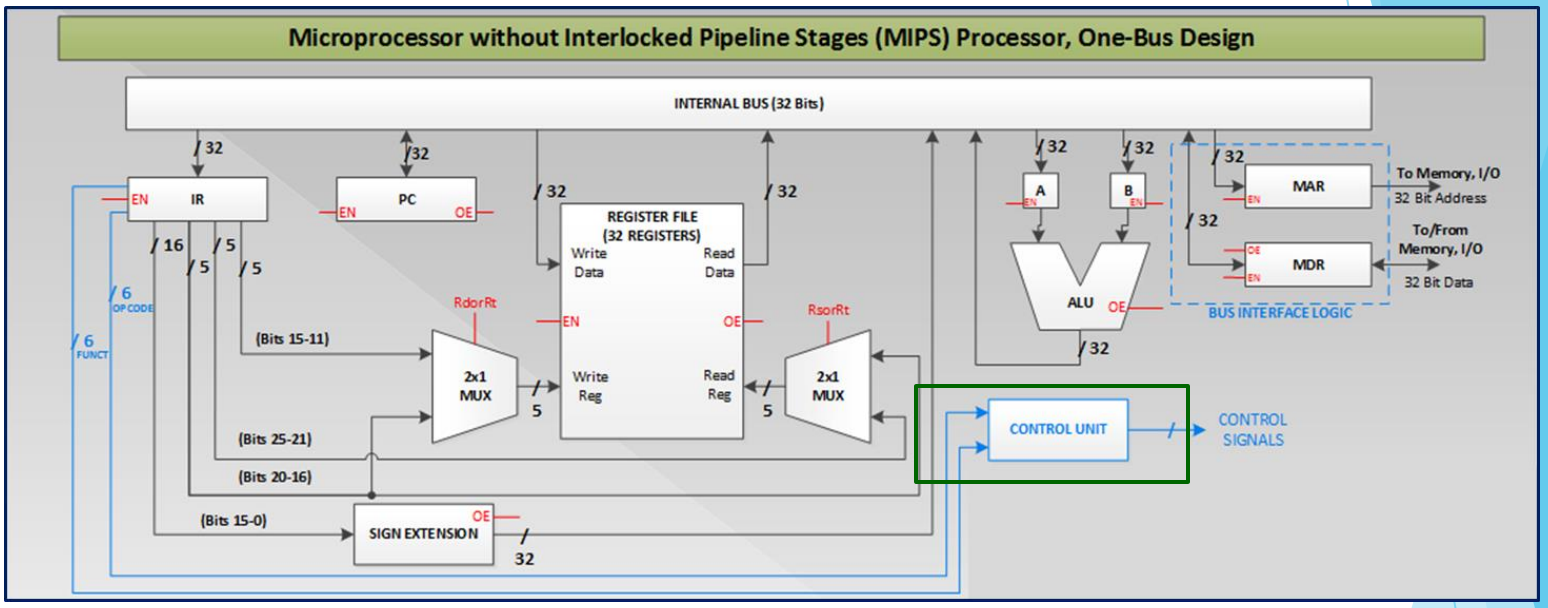
- Sign-extends the 16-bit immediate value to 32 bits.
  - Extends the sign bit from bit 15 to bit 31.
  - Maintains the sign of the immediate operand (in the MIPS I-type instruction).
- The ALU is combinational logic.
  - Does not require a clock.
  - Does not store information.

Control Unit



- Decodes the instruction that is in the Instruction Register (IR).
  - *Opcode* field (for all instructions).
  - *Funct* field (for R-type instructions only).
- Generates the control signals (for the datapath).
  - Asserts the appropriate control signals for the instruction being executed.

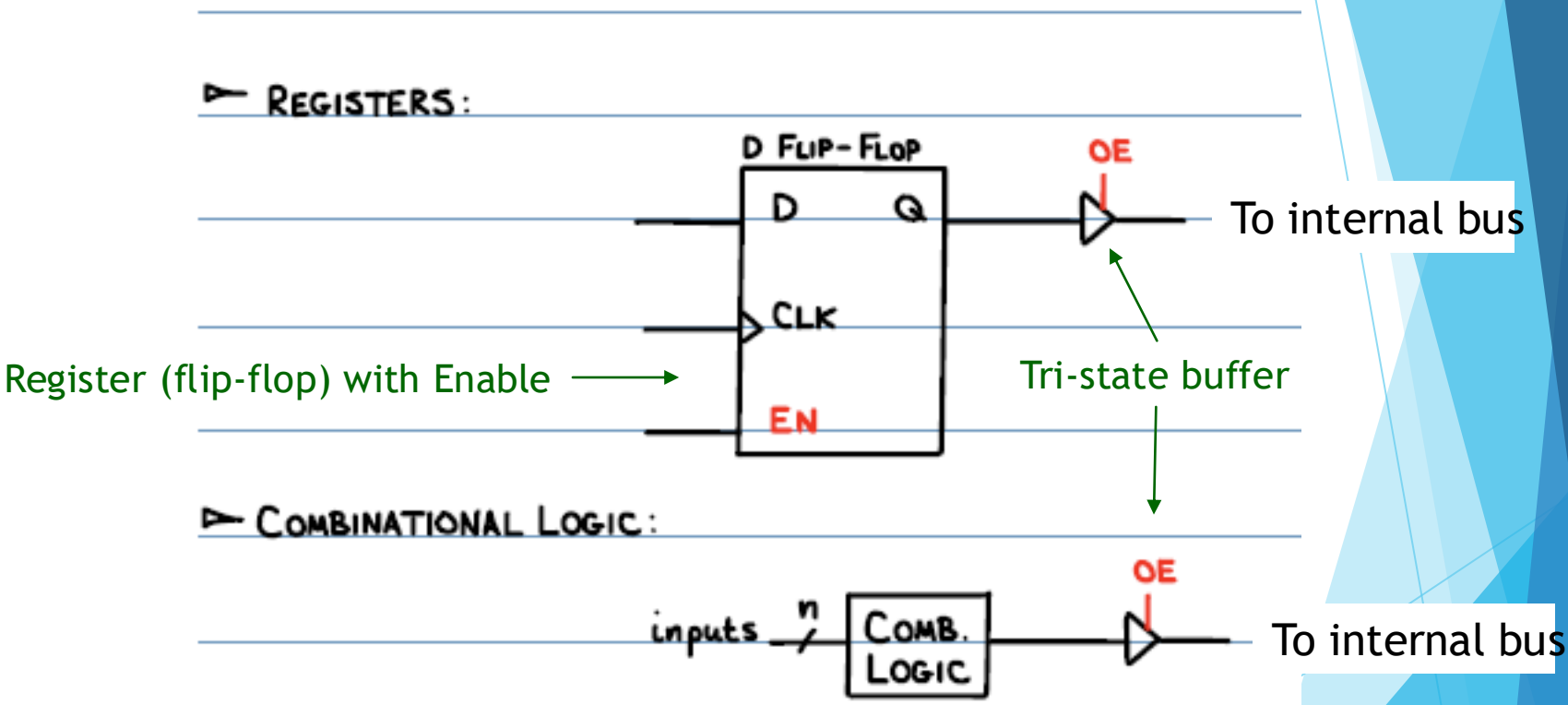
MIPS (One-bus): Control Unit



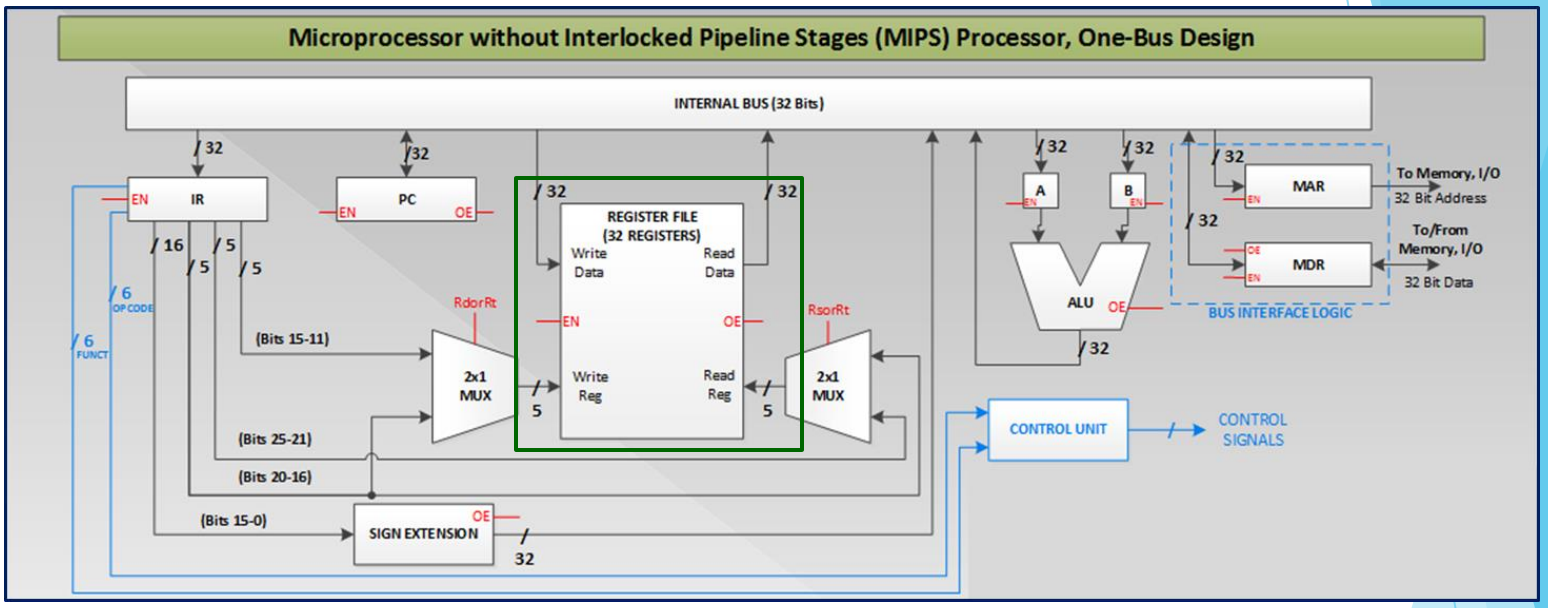
- Decodes the instruction that is in the Instruction Register (IR).
  - *Opcode* field (for all instructions).
  - *Funct* field (for R-type instructions only).
- Generates the control signals (for the datapath).
  - Asserts the appropriate control signals for the instruction being executed.

Control signals can be active-high or active-low.

EN and OE Control Signals

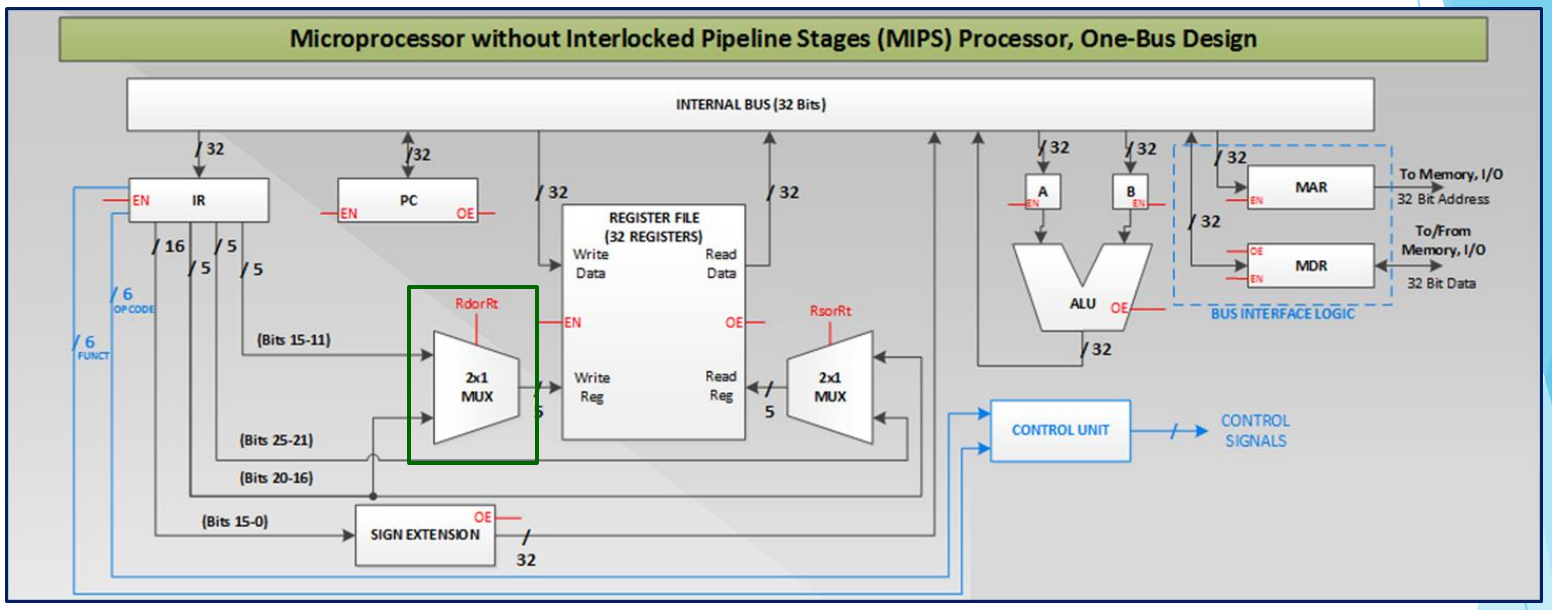


Register File



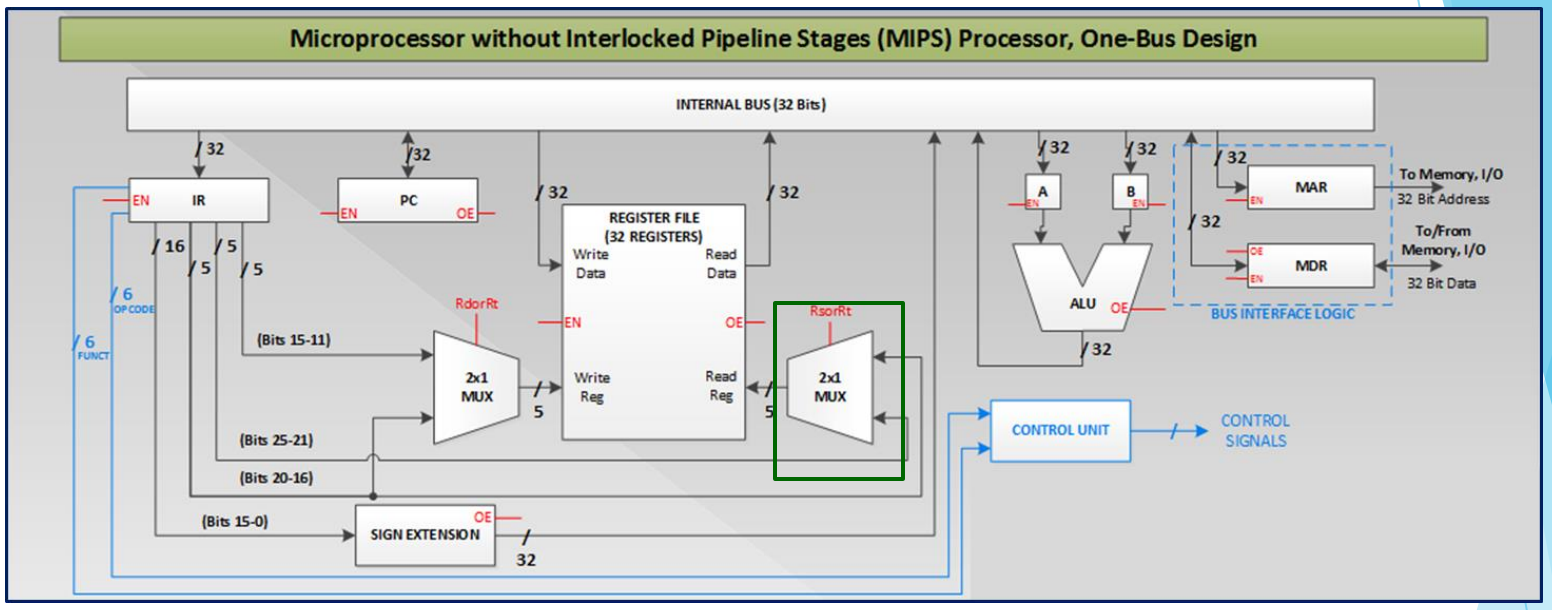
Datapath Component	Signal	Description
Register File	RF_{EN}	When asserted, RF inputs data from CPU bus.
	RF_{OE}	When asserted, RF tri-state buffer is enabled and RF outputs data to CPU bus.

Write Register Multiplexer



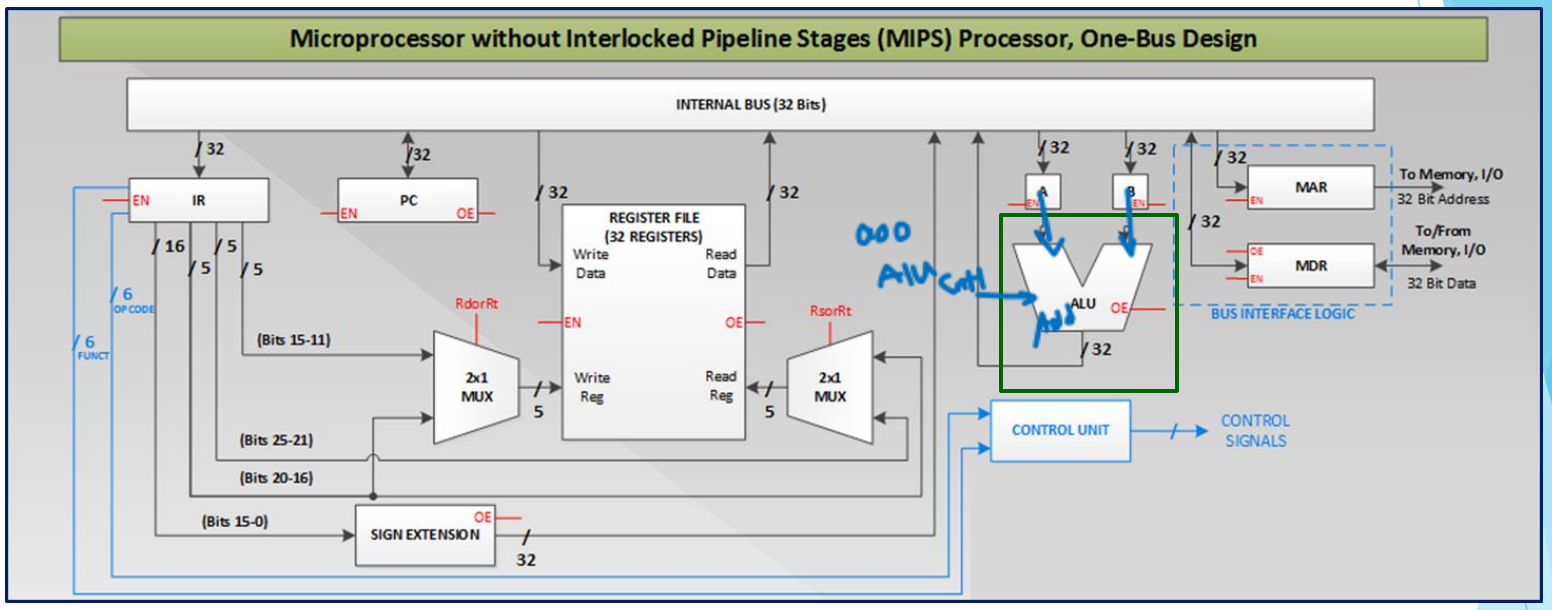
Datapath Component	Signal	Description
Write register Mux	RDorRT	RDorRT = 0: Use Rd field to specify destination reg. RDorRT = 1: Use Rt field to specify destination reg.

Read Register Multiplexer



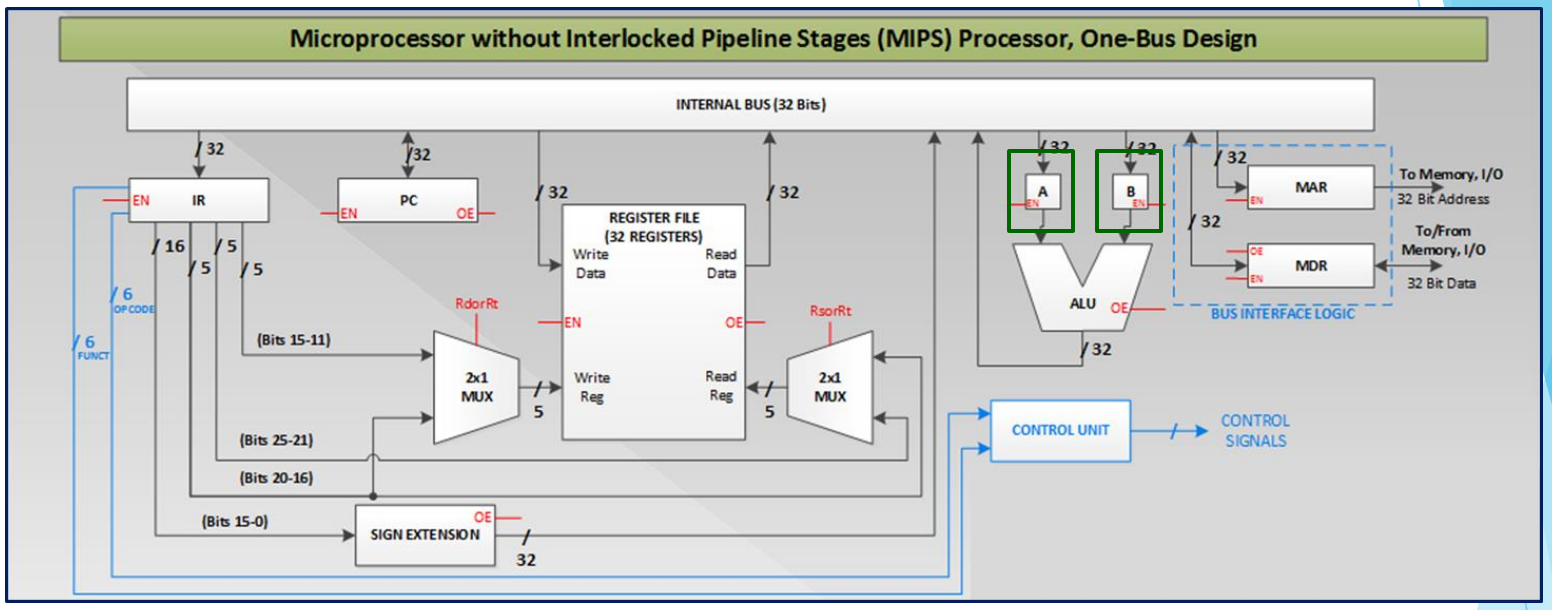
Datapath Component	Signal	Description
Read register Mux	RSorRT	RSorRT = 0: Use Rs field to specify source register. RSorRT = 1: Use Rt field to specify source register.

Arithmetic and Logic Unit (ALU)



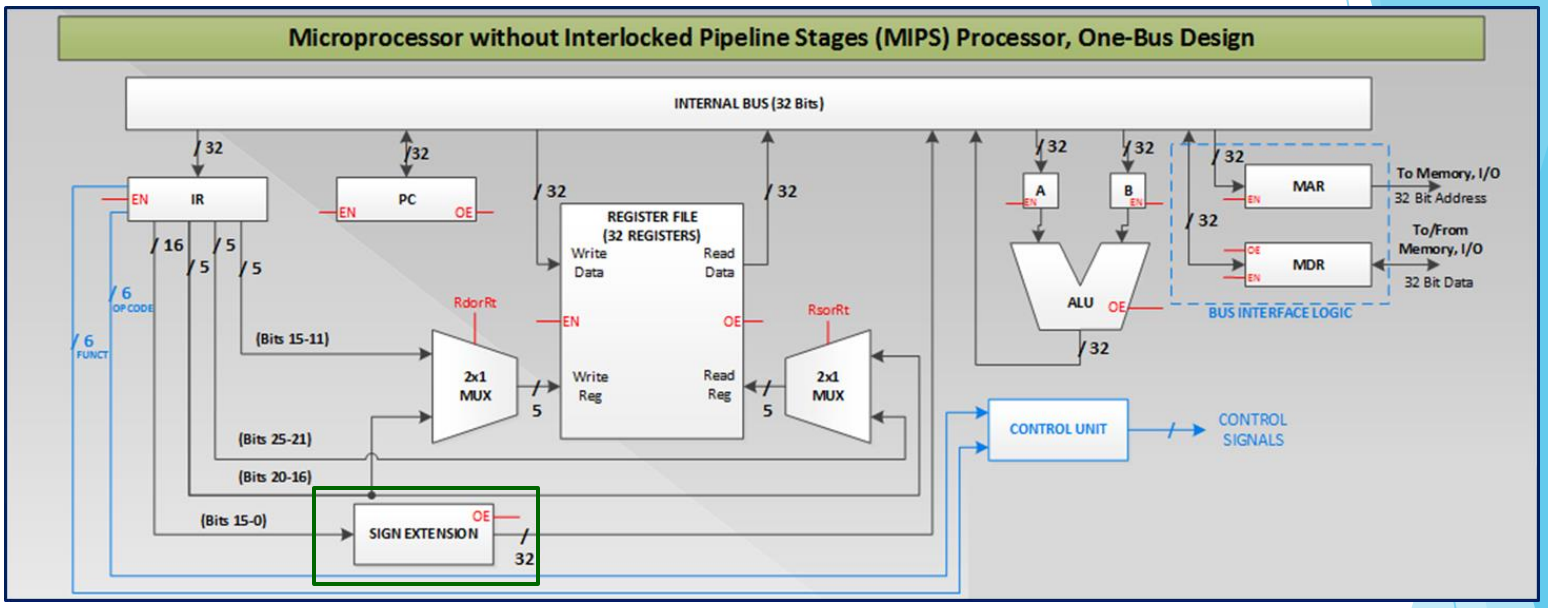
Datapath Component	Signal	Description
ALU	ALU_{OE}	When asserted, ALU tri-state buffer is enabled and ALU outputs data to CPU bus.
	ALU_{CNTL}	Controls the operation performed by the ALU.

Temporary Registers



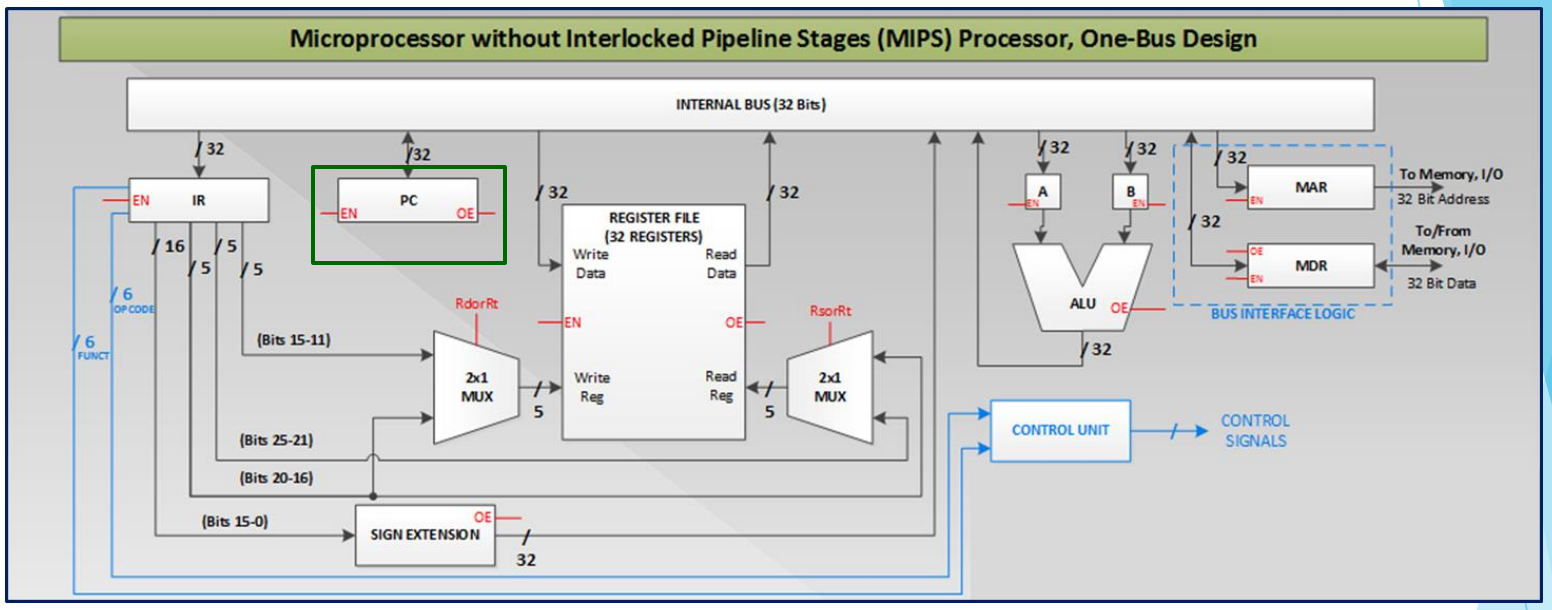
Datapath Component	Signal	Description
Register A	A_{EN}	When asserted, Reg A inputs data from CPU bus.
Register B	B_{EN}	When asserted, Reg B inputs data from CPU bus.

Sign-Extension Unit (SEU)



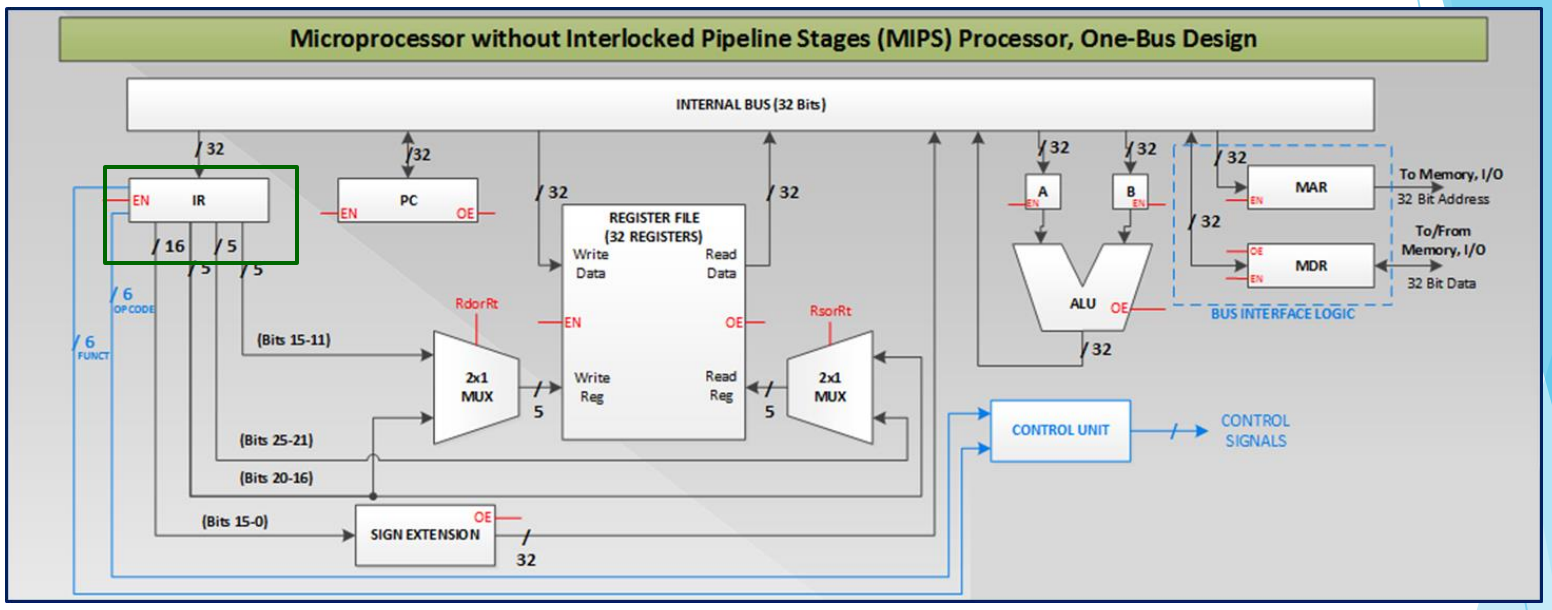
Datapath Component	Signal	Description
Sign-extension Unit	SEU_{OE}	When asserted, SEU tri-state buffer is enabled and SEU outputs data to CPU bus.

Program Counter



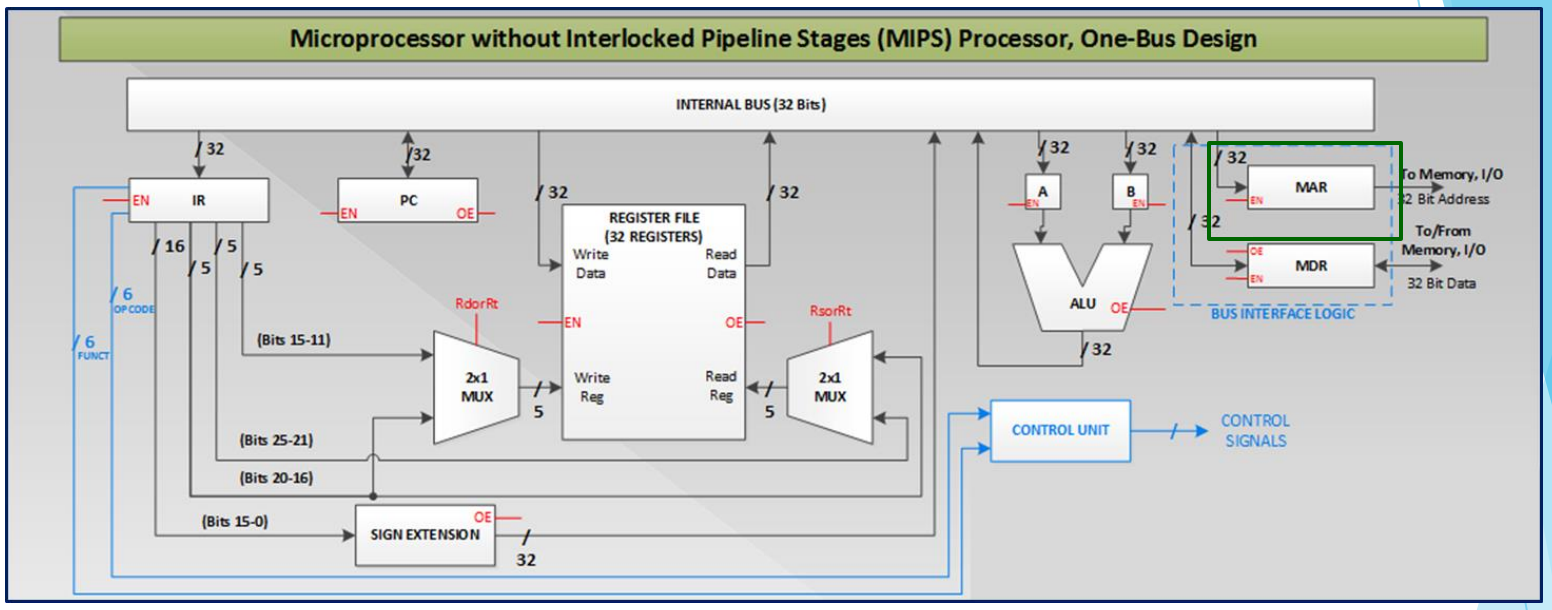
Datapath Component	Signal	Description
Program Counter	PC_{EN}	When asserted, PC inputs data from CPU bus.
	PC_{OE}	When asserted, PC tri-state buffer is enabled and PC outputs data to CPU bus.

Instruction Register



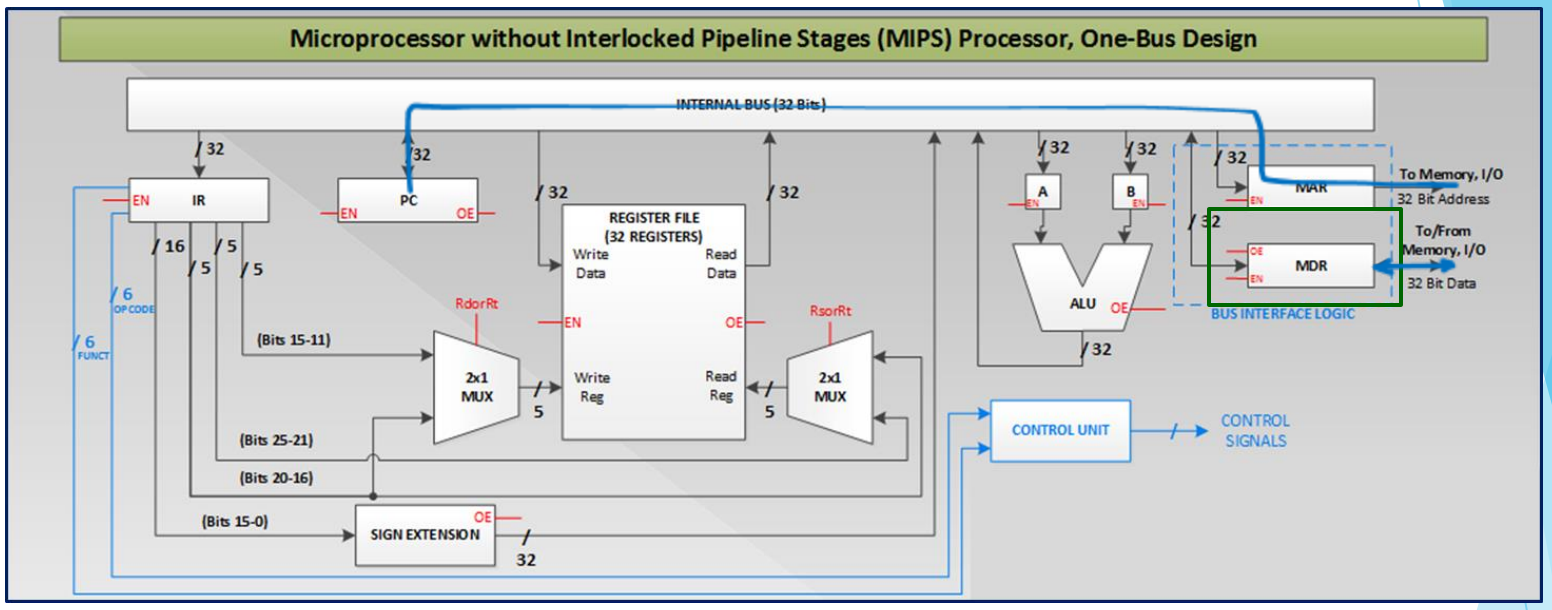
Datapath Component	Signal	Description
Instruction Register	IR_{EN}	When asserted, IR inputs data from CPU bus.

Memory Address Register (MAR)



Datapath Component	Signal	Description
Memory Address Reg	MAR_{EN}	When asserted, MAR inputs data from CPU bus.

Memory Data Register (MDR)



Datapath Component	Signal	Description
Memory Data Reg	MDR_{EN}	When asserted, MDR inputs data from CPU bus.
	MDR_{OE}	When asserted, MDR tri-state buffer is enabled and MDR outputs data to CPU bus.

MIPS (One-bus): Control Signals

Datapath Component	Signal	Description
Program Counter	PC_{EN}	When asserted, PC inputs data from CPU bus.
	PC_{OE}	When asserted, PC tri-state buffer is enabled and PC outputs data to CPU bus.
Instruction Register	IR_{EN}	When asserted, IR inputs data from CPU bus.
Register A	A_{EN}	When asserted, Reg A inputs data from CPU bus.
Register B	B_{EN}	When asserted, Reg B inputs data from CPU bus.
Memory Address Reg	MAR_{EN}	When asserted, MAR inputs data from CPU bus.
Memory Data Reg	MDR_{EN}	When asserted, MDR inputs data from CPU bus.
	MDR_{OE}	When asserted, MDR tri-state buffer is enabled and MDR outputs data to CPU bus.

- PC and MDR require bi-directional data transfer via the internal CPU bus.
- All other registers only need to read (input) from the internal CPU bus.

MIPS (One-bus): Control Signals

Datapath Component	Signal	Description
Register File	RF_{EN}	When asserted, RF inputs data from CPU bus.
	RF_{OE}	When asserted, RF tri-state buffer is enabled and RF outputs data to CPU bus.
Read register Mux	RSorRT	RSorRT = 0: Use Rs field to specify source register. RSorRT = 1: Use Rt field to specify source register.
Write register Mux	RDorRT	RDorRT = 0: Use Rd field to specify destination reg. RDorRT = 1: Use Rt field to specify destination reg.
Sign-extension Unit	SEU_{OE}	When asserted, SEU tri-state buffer is enabled and SEU outputs data to CPU bus.
ALU	ALU_{OE}	When asserted, ALU tri-state buffer is enabled and ALU outputs data to CPU bus.
	ALU_{CNTL}	Controls the operation performed by the ALU.

MIPS (One-bus): ALU and Memory

ALU Control signals

ALU_{CNTL}	Operation
0000	AND
0001	OR
0010	Add
0110	Subtract
0111	Set on Less Than
0011	Increment by 4

A maximum of 16 operations can be defined for the ALU.
(using a 4-bit control signal)

Memory Control signals

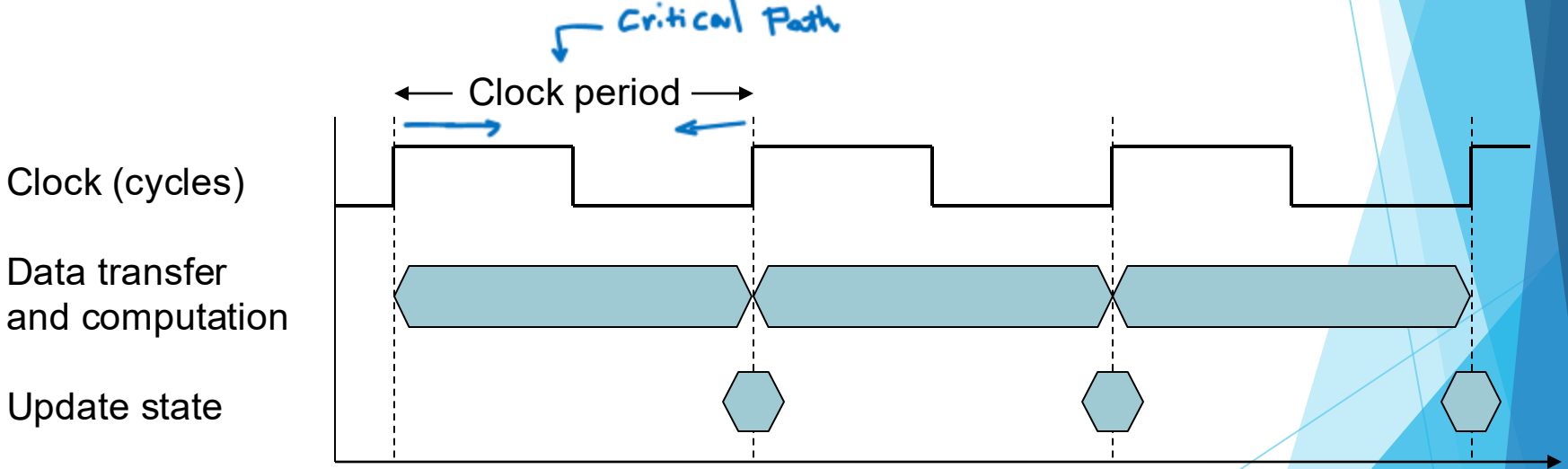
Signal	Operation
Read	When asserted, data is transferred ***from*** memory.
Write	When asserted, data is transferred ***to*** memory.

Both cannot be asserted at the same time.

Signal	Operation
R/W'	When low (0), data is transferred ***to*** memory. (write)
	When high (1), data is transferred ***from*** memory. (read)

Processor Clock Cycles

- Processor operation is governed by a constant-rate clock.
- Instruction cycle is, thus, controlled by the clock as well.
  - Each step in the instruction cycle requires one clock cycle.
  - Instruction processing described using register transfer language (RTL).

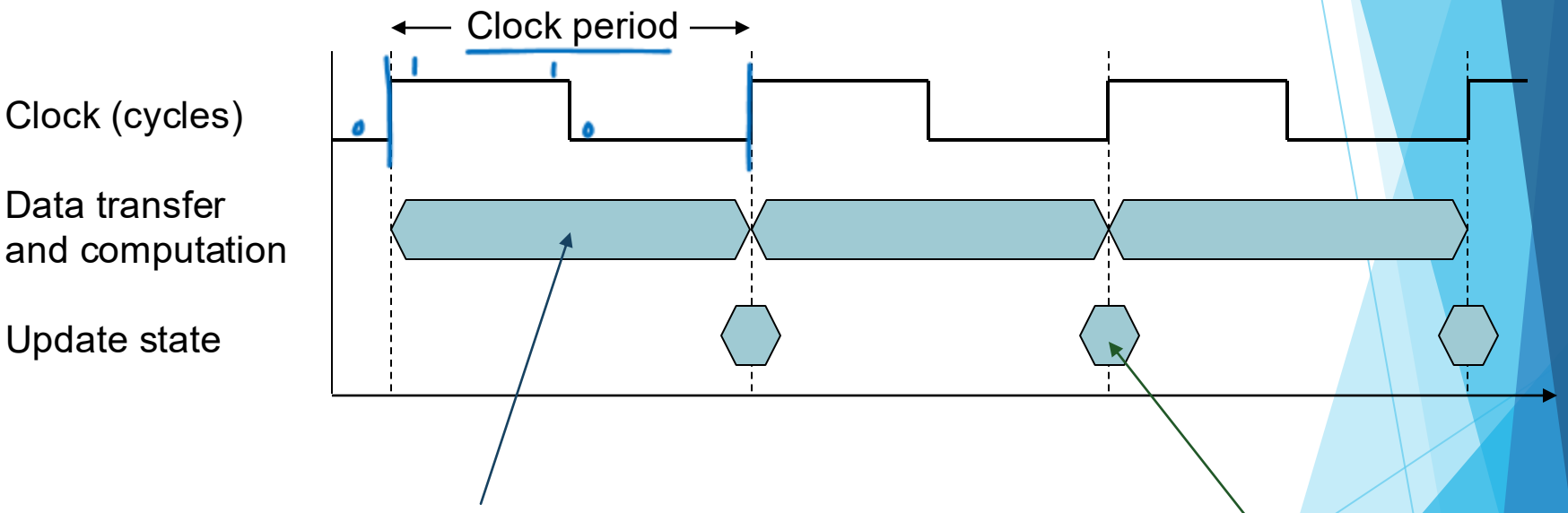


T_{CLK} = clock period (clock cycle time)

$\frac{1}{T_{CLK}} = F_{CLK}$ = clock frequency (clock rate)

Processor Clock Cycles

- Operation of the CPU is governed by a constant-rate clock.



- Data is transferred from one register to another on the internal bus.
- Calculations are carried out by the ALU.

Registers are updated on the rising-edge of the clock.

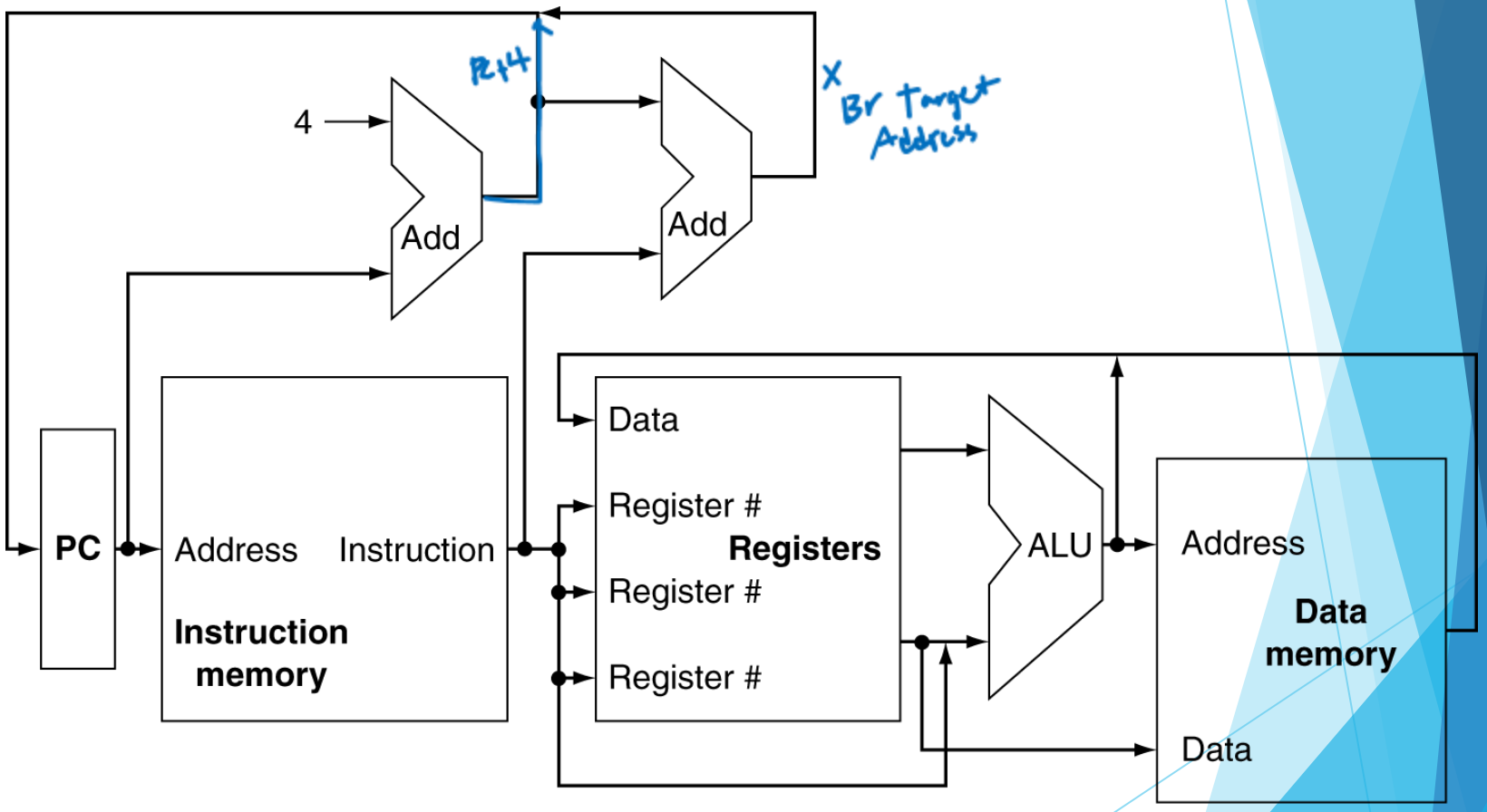
Building MIPS Datapath

Instruction Execution

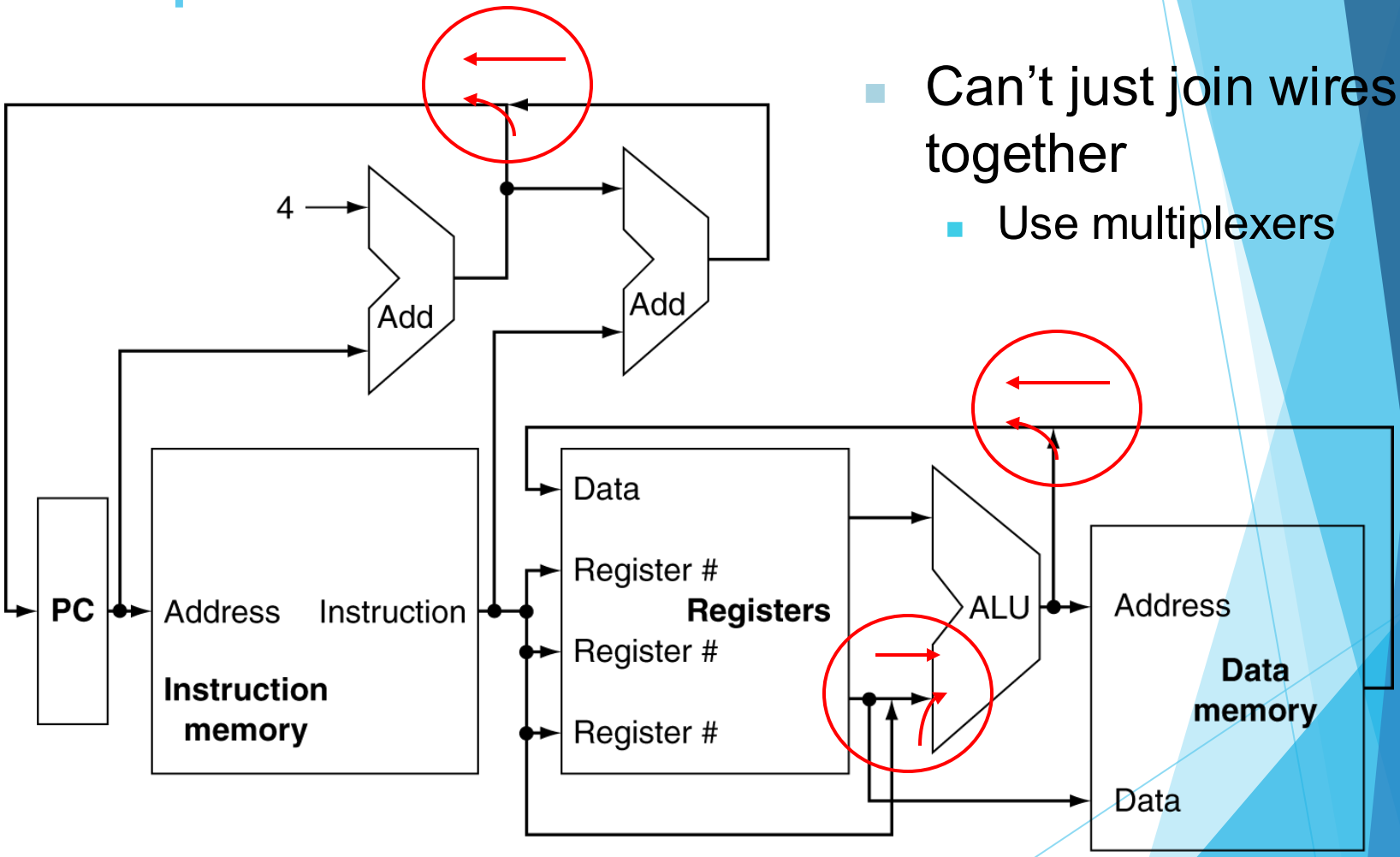
- PC → instruction memory, fetch instruction
- Register numbers → register file, read registers
- Depending on instruction class
  - Use ALU to calculate
    - Arithmetic result
    - Memory address for load/store
    - Branch target address
  - Access data memory for load/store
  - PC ← target address or PC + 4

Sequential order

CPU Overview

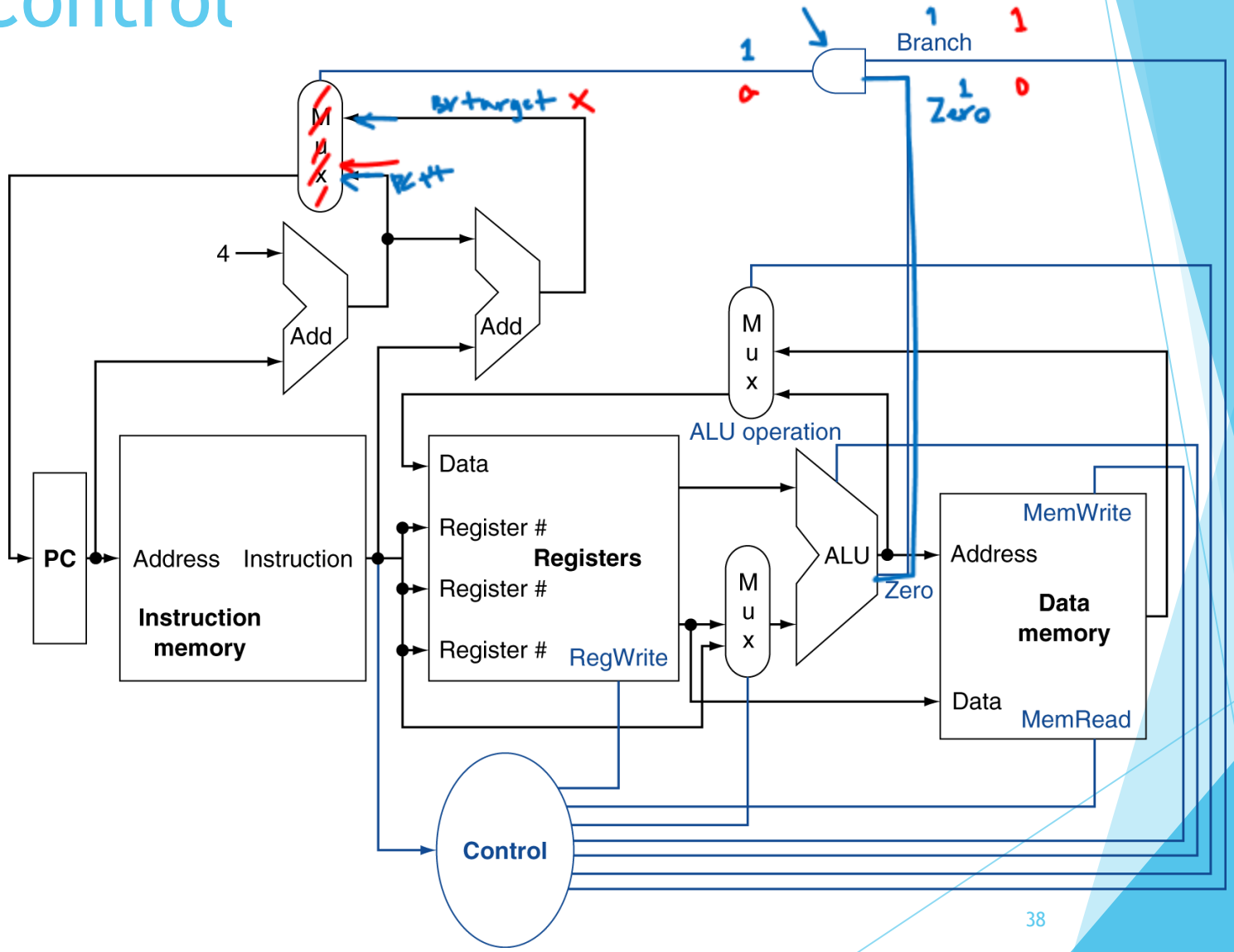


Multiplexers

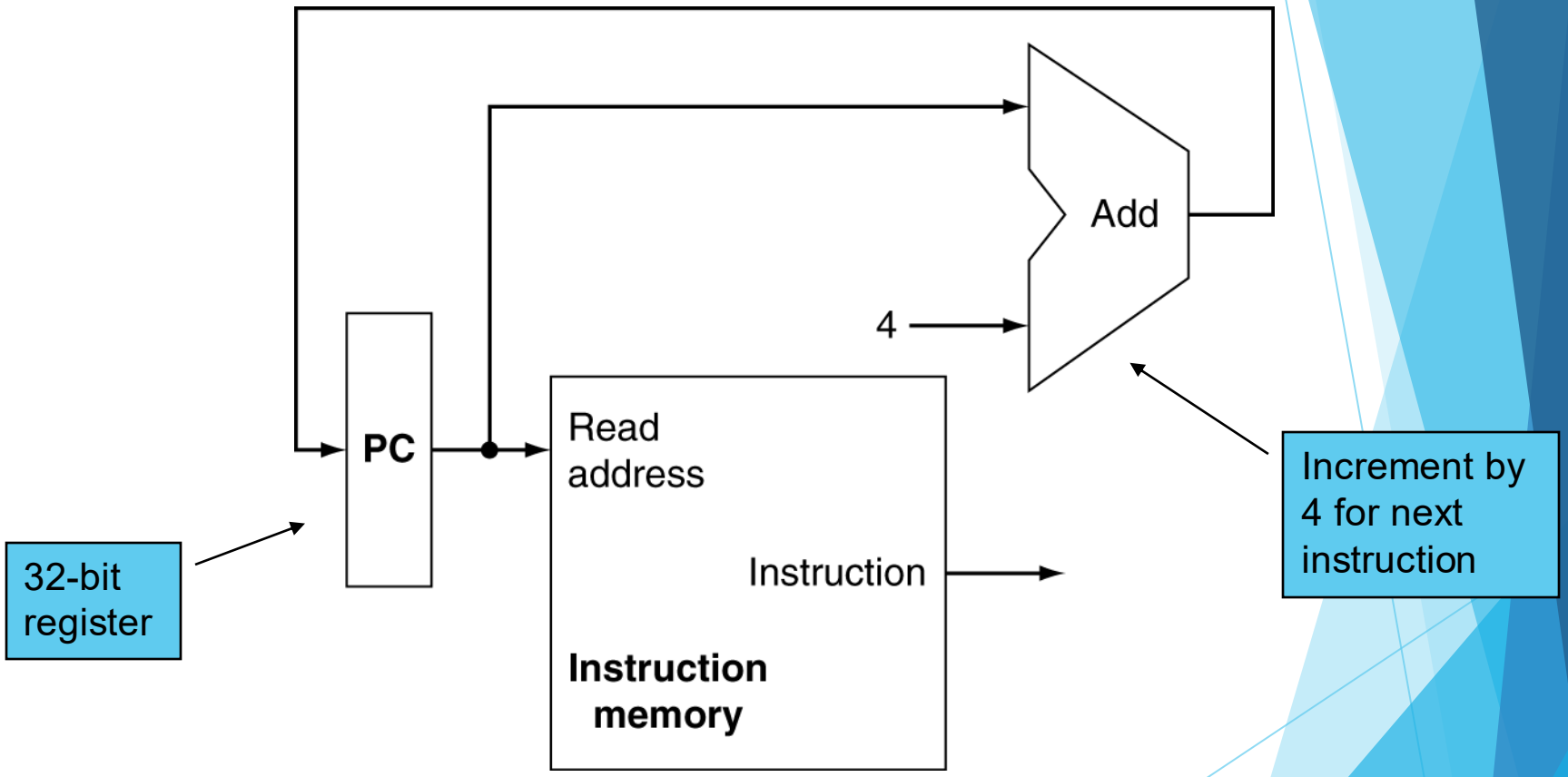


- Can't just join wires together
  - Use multiplexers

Control

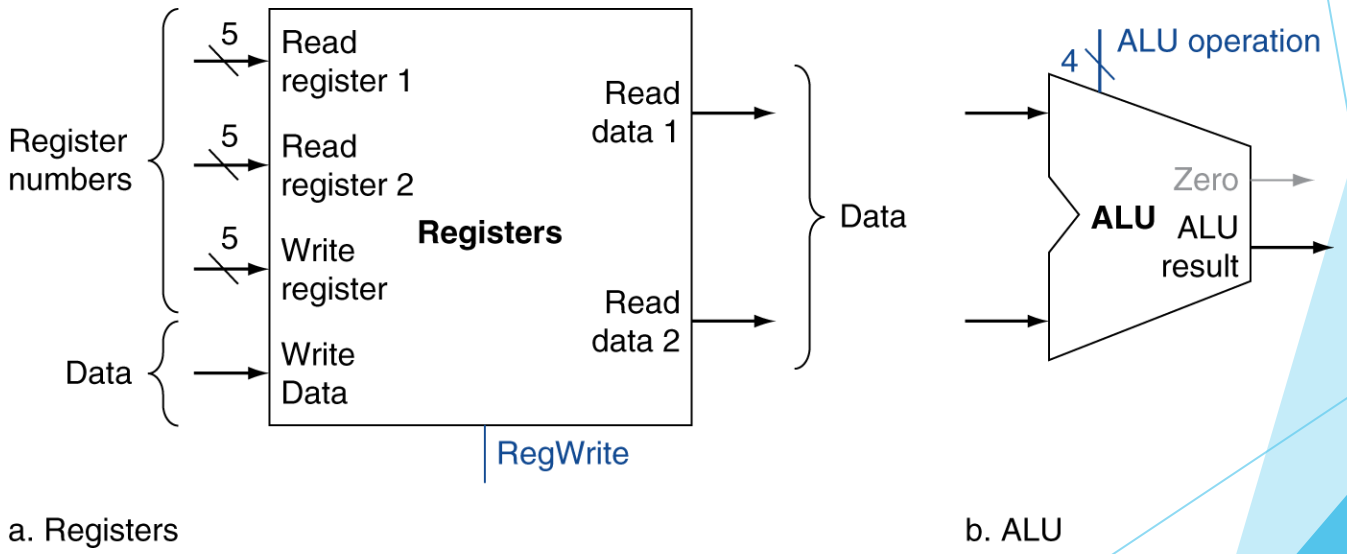

Branch
Zero
M
u
x
4
Add
Add
ALU operation
PC
Address
Instruction
Instruction memory
Data
Register #
Registers
Register #
Register #
RegWrite
ALU
Zero
MemWrite
Address
Data memory
Data
MemRead
Control

Instruction Fetch



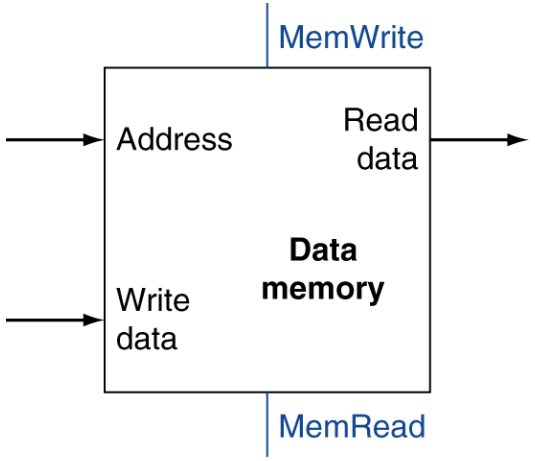
R-Format Instructions

- Read two register operands
- Perform arithmetic/logical operation
- Write register result

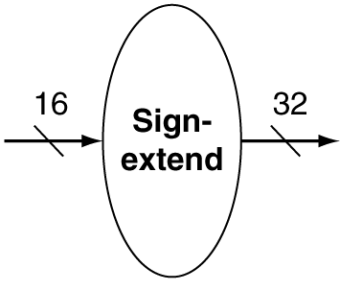


Load/Store Instructions

- Read register operands
- Calculate address using 16-bit offset
  - Use ALU, but sign-extend offset
- Load: Read memory and update register
- Store: Write register value to memory



a. Data memory unit

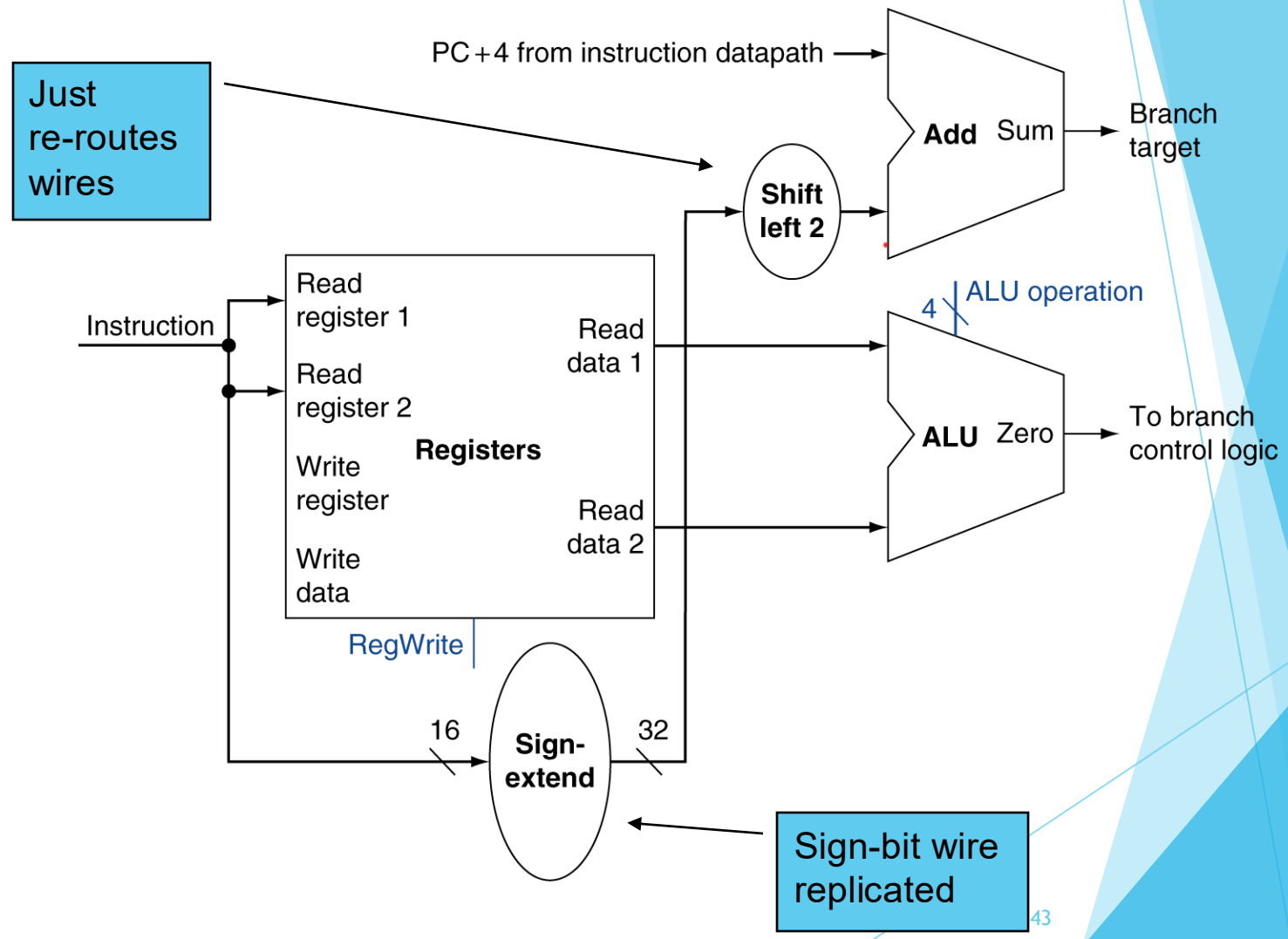


b. Sign extension unit

Branch Instructions

- Read register operands
- Compare operands
  - Use ALU, subtract and check Zero output
- Calculate target address
  - Sign-extend displacement 16 bit Imm SEN → 32 bits
  - Shift left 2 places (word displacement)
  - Add to PC + 4
    - Already calculated by instruction fetch

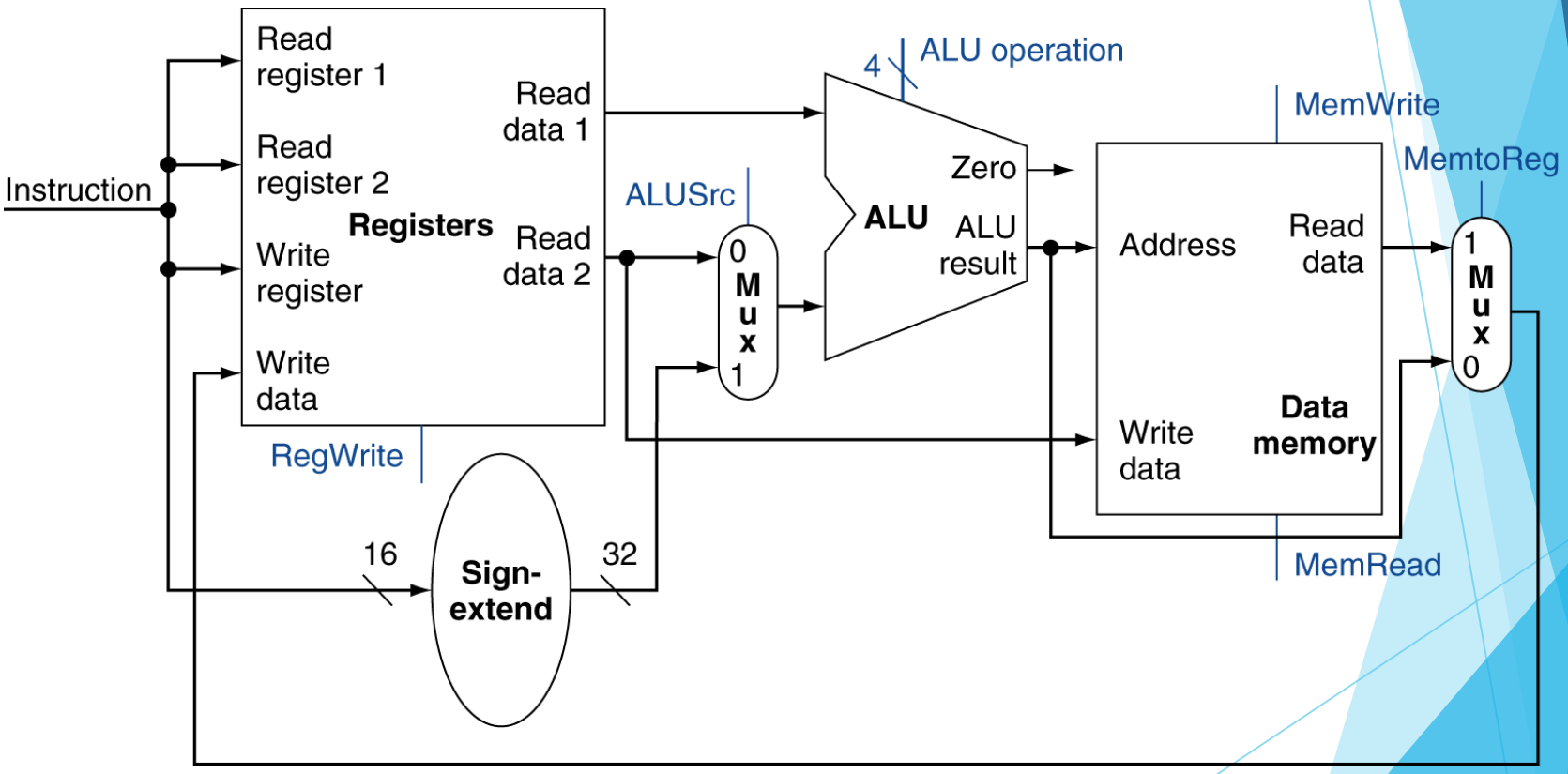
Branch Instructions



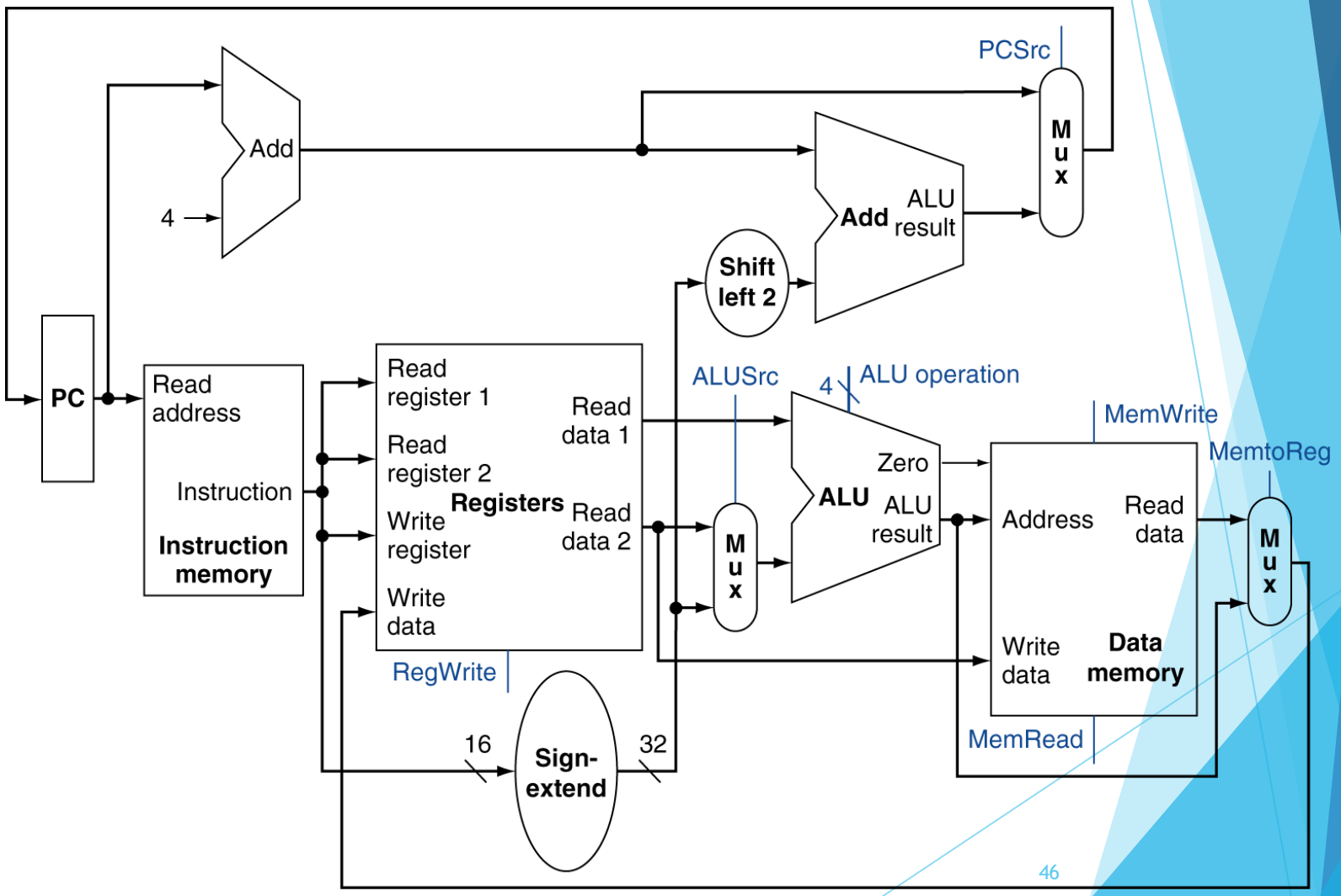
Composing the Elements

- First-cut data path does an instruction in one clock cycle
  - Each datapath element can only do one function at a time
  - Hence, we need separate instruction and data memories
- Use multiplexers where alternate data sources are used for different instructions

R-Type/Load/Store Datapath



Full Datapath



ALU Control

- ALU used for
  - Load/Store: F = add
  - Branch: F = subtract
  - R-type: F depends on funct field

ALU control	Function
0000	AND
0001	OR
0010	add
0110	subtract
0111	set-on-less-than
1100	NOR

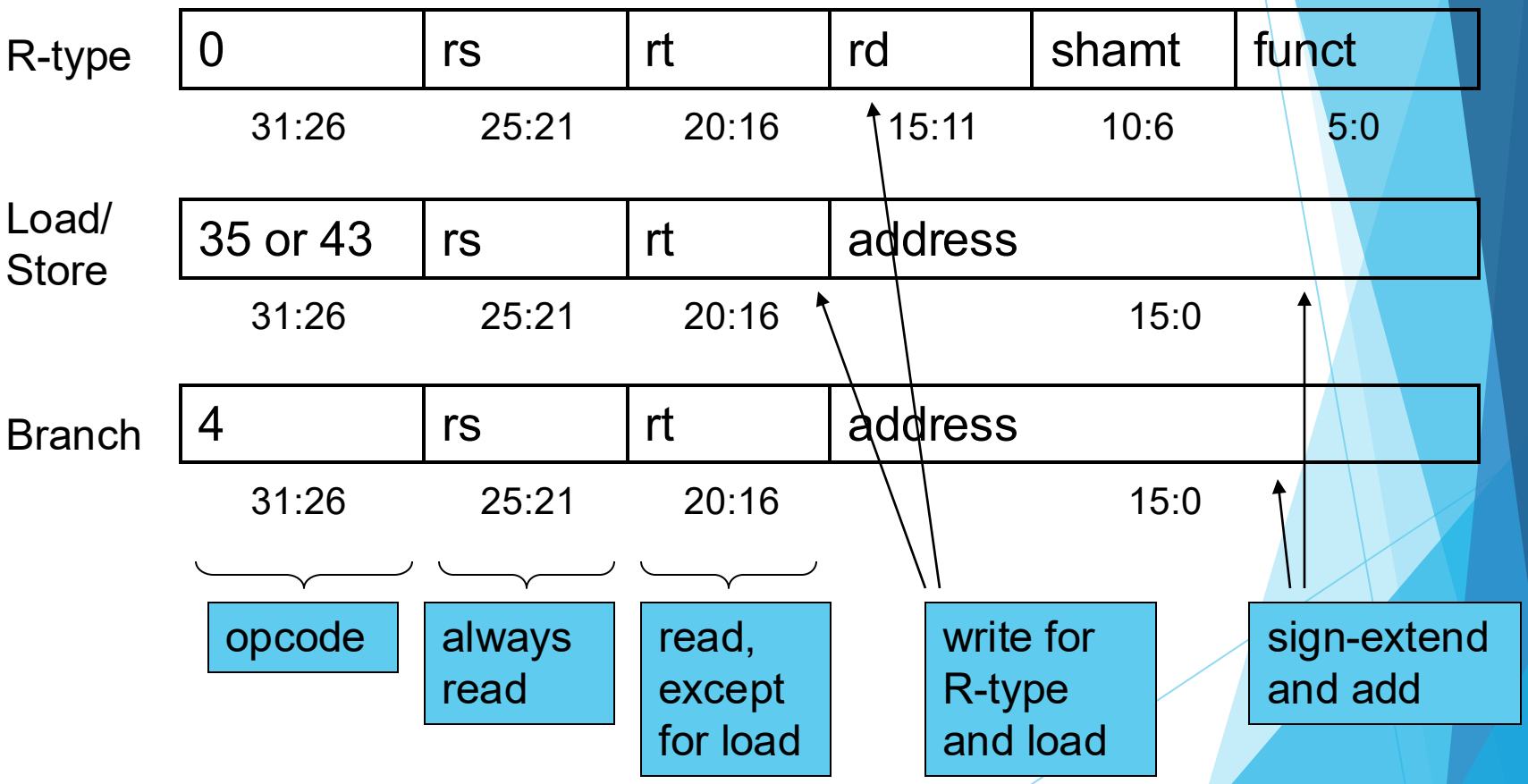
ALU Control

- Assume 2-bit ALUOp derived from opcode
  - Combinational logic derives ALU control

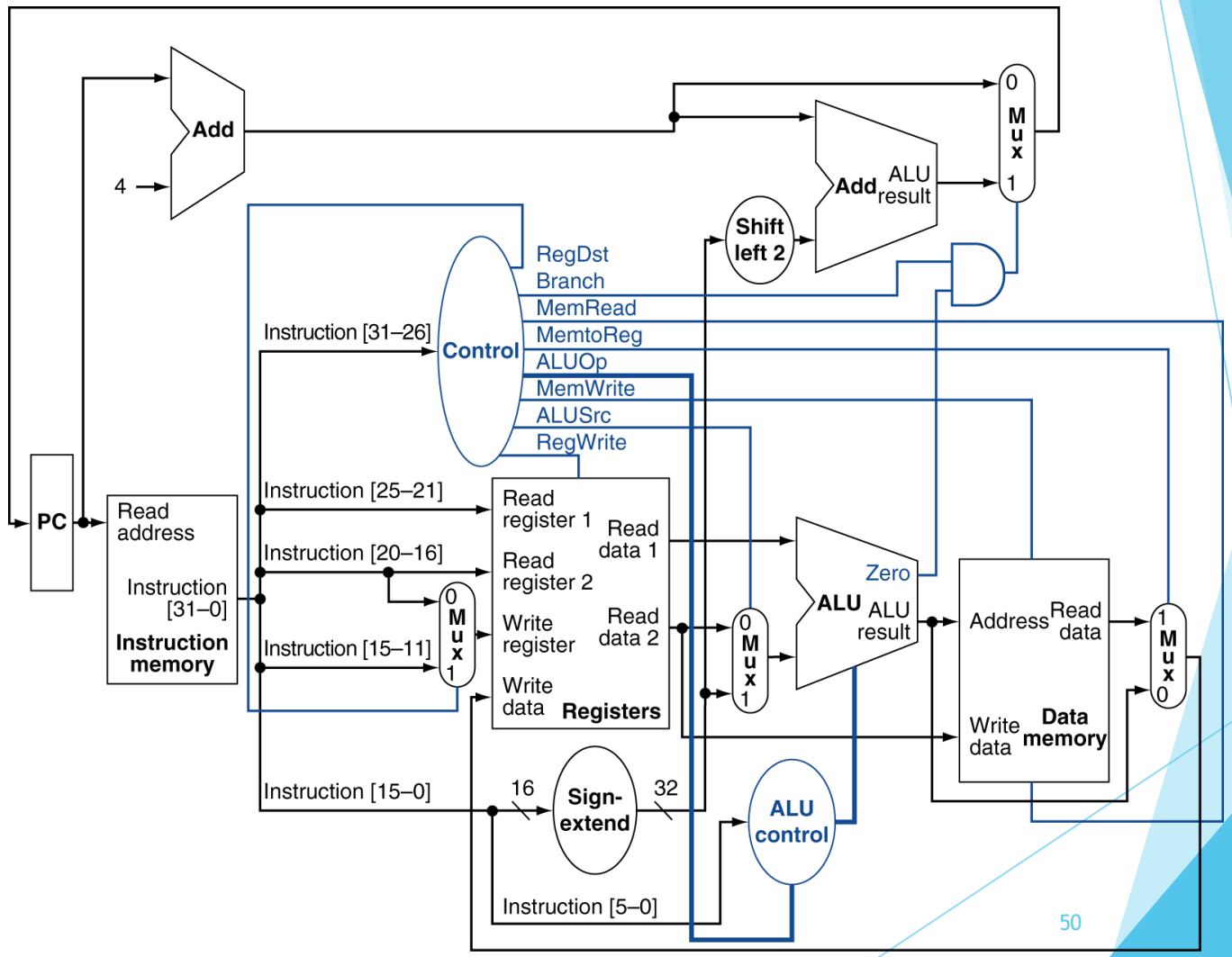
opcode	ALUOp	Operation	funct	ALU function	ALU control
lw	00	load word	XXXXXX	add	0010
sw	00	store word	XXXXXX	add	0010
beq	01	branch equal	XXXXXX	subtract	0110
R-type	10	add	100000	add	0010
		subtract	100010	subtract	0110
		AND	100100	AND	0000
		OR	100101	OR	0001
		set-on-less-than	101010	set-on-less-than	0111

The Main Control Unit

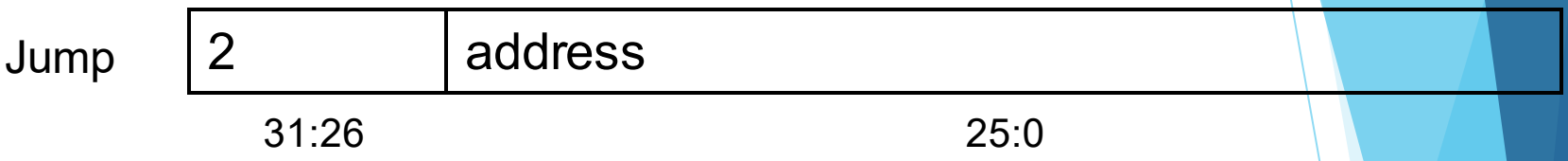
- Control signals derived from instruction



Datapath With Control



Implementing Jumps



- Jump uses word address
- Update PC with concatenation of
  - Top 4 bits of old PC
  - 26-bit jump address
  - 00
- Need an extra control signal decoded from opcode

Datapath With Jumps Added

